DIAGNOSTIC BENCHMARK / RESEARCH MANUSCRIPT

# Silent Acceptance: A Benchmark of Specification Gates and Extrapolation Failures in Physical Model Assessment

Detection, false alarms, and joint release for prespecified power-law candidates.

Fixed external regions, repeated calibration, and structurally grouped confirmation across the full sample-size and noise grid.

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# Silent Acceptance: A Benchmark of Specification Gates and Extrapolation Failures in Physical Model Assessment


## Abstract

Specification gates challenge candidate equations, but a non-rejecting record provides protection only against departures the selected policy can detect. We evaluate power-law candidates using residual-permutation curvature, RESET variants, quadratic-rival comparisons, and a fixed radial-basis-function rival. We connect fit screening, independently calibrated diagnostic policies, and error over external regions fixed before training data are sampled.

The historical benchmark identifies directional blind spots and numerical sensitivity; its single-probe joint-release result is retained as a separate descriptive analysis. New validation first uses fresh data for 20 independent reference/calibration pairs on the known equations. An H0-only robust affine score removes artificial empirical-CDF tail saturation without eliminating intrinsic statistic ties. The main fixed-domain evaluation freezes structural groups, positive valid regions, grids, weights, and outcomes before sampling. It separates historically evaluated equations, source-derived physical tasks, designed mechanisms, and reserve groups withheld from study outcomes; calibration and confirmation nulls test design transfer within the same monomial family.

The complete fixed-domain study contains 246,240 datasets across three sample sizes and four noise levels. At 3% log-noise, the quadratic AICc gate releases 41.59%, 21.88%, and 9.36% of screened-dangerous wrong fits at n=50, 200, and 1000, versus 64.41%, 32.99%, and 15.65% for the A/E/F panel at the same 5% calibration budget. These primary dangerous cases come from designed mechanisms. The fit screen intercepts all observed dangerous physical-source fits, leaving no screened-dangerous case to estimate the gates' conditional release rate in this six-source-task design.

Equal calibration budgets do not imply equal test false-rejection rates. Group and calibration-round uncertainty remains conditional on the registered library and realized test bank. The evidence supports measuring residual release and correct-model losses together; it does not establish universal gate rankings, broad physical-law coverage, or safety of expressions selected by a discovery algorithm.


| **Candidate fit**<br>Prespecified power law | **Fit screen**<br>$R^2 \geq 0.99$ | **Gate panel**<br>No rejection or decline | **Registered probe**<br>Fixed-region or legacy error >10%? |
|---|---|---|---|

Figure 1 - Evaluation stages. Candidate fitting and the fit screen define the screened population; valid gate decisions define release; the registered extrapolation probe measures its consequence.

# 1 Introduction

A candidate equation is useful only to the extent that the available evidence supports its intended use. Prediction error describes agreement on sampled inputs; specification diagnostics ask whether selected departures from an assumed form remain detectable. A workflow may fit a candidate, require a high training score, run diagnostics, and release models that trigger no alarm. The practical question is what errors that complete rule removes and what errors it still releases.

The distinction is concrete in a controlled friction example. A power law fitted to Prandtl-Karman data reaches $R^2 = 0.99$, passes cross-validation and an information-criterion comparison, yet incurs about -20% error one decade beyond training. The four specification tests studied here detect that synthetic departure at power $\geq 0.99$. A fit-only failure can therefore be detectable, while other mismatch directions survive the same checks.

We assess prespecified power-law candidates in log coordinates. Permutation curvature compresses departure from log-linearity into one quadratic direction; RESET examines powers of fitted values or regressors; rival models add quadratic features or a frozen RBF residual correction. Each construction has a sensitivity geometry. Its usefulness also depends on false alarms, numerical stability, the fit screen, and the policy by which several statistics are combined.

The historical S1-S7 suites map these effects using controlled alternatives, physics-derived equations, family viability, rival comparisons, and observational compatibility probes. They also expose the limits of a cached endpoint analysis: the external coordinate depends on the training sample, threshold crossings concentrate in two equations, and individual reference thresholds do not control a joint panel's independent-test cost. We retain those results as historical evidence rather than treating them as the final policy comparison.

Two prospective validation layers address these limitations. P0-1 v2 uses wholly fresh samples and 20 independent reference/ calibration pairs, preserving the known equation identities and legacy endpoint. It separates calibration sensitivity from task composition and fixes an artificial tail ceiling in the initial score mapping. P0-2 then freezes task grouping, valid external regions, evaluation grids, and a region-level primary outcome before training sampling. Historical tasks and algebraic near-twins remain in the development pool. Newly evaluated physical-source tasks and designed positive mechanisms are reported separately; variants and parameter windows remain inside their structural groups.

Three findings organize the study. First, sensitivity follows the departure direction, so a single diagnostic ranking does not describe all alternatives. Second, joint policy calibration changes the value of adding a gate: a panel's threshold must account for the complete policy, and equal calibration targets still allow different independent-test H0 costs. Third, gate silence and extrapolation consequences must be linked on the same fits. Fixed-region release and whole-workflow correct-model retention make that link explicit.

We make three contributions:

1. **A benchmark with explicit decision populations.** Form labels, fit screening, alarms, declines, release, and error outcomes are registered separately. Historical single-probe analyses and new fixed-region analyses keep their distinct denominators.

2. **A paired comparison of independently calibrated policies.** Single gates and prespecified panels share data and an H0-only joint calibration procedure. Repeated reference/calibration draws quantify policy sensitivity without counting repeated decisions on the shared test bank as independent observations.

3. **A fixed-domain structural-group assessment.** Regions, grids, weights, and truth are frozen before training draws. Near-twins do not cross evaluation roles. Structural-group and calibration- round resampling distinguish equation coverage from simulation volume, while physical-source and designed-mechanism evidence remain identifiable.

Symbolic-regression benchmarks evaluate predictive quality, expression complexity, and symbolic or structural recovery [5, 6]. Our main experimental object is the behavior of attached specification policies under a fixed candidate family. Validation of expressions actually selected by a discovery algorithm remains a separate requirement. The proposed expansion of physical coverage in Section 6 has not yet generated experimental outcomes.

Sections 3 and 4 define the protocols and results. Section 5 discusses diagnostic and workflow implications; Section 6 states the remaining coverage and selection limits. Section 7 and the appendices provide provenance, correction history, and the boundary between completed evidence and registered future work.

# 2 Related Work

**Specification testing and robust inference.** Ramsey's RESET [1] tests powers of fitted values; regressor-power variants examine other directions of misspecification. Heteroskedasticity-consistent covariance estimation [2] and HC3 leverage adjustment [3, 4] address inference under nonconstant error variance. We compare these established tools on log-transformed, physics-derived designs. The numerical audit in Section 5.2 and Appendix G concerns implementation of the HC3-Wald calculation; the statistical correction and its numerical realization must be distinguished.

**Model comparison as a diagnostic.** Information criteria [12] and cross-validation [13] can compare a candidate with a richer rival. Our S5 comparison freezes a quadratic rival and reports both native and H0-reference-calibrated policies. This establishes a practical alternative to the single-direction curvature statistic. Its comparison with A is interpreted using the measured error budgets in Section 4.7.

**Resampling and transformed coordinates.** Wild-bootstrap methods [10, 11] provide tools for inference under heteroskedasticity. Residual permutation relies on exchangeability-related conditions. The family-viability experiment examines how these choices behave after linearizing transformations: replacing permutation with a wild null substantially reduces the observed rational-coordinate distortion. The reported calibration evidence is empirical for the tested designs.

**Symbolic-regression evaluation.** SRBench [5] evaluates predictive accuracy, expression complexity, and symbolic recovery. SRSD [6] revises physically motivated sampling configurations and evaluates structural discrepancy using normalized edit distance. We reuse SRSD configurations to assess diagnostic decisions for prespecified power laws. The measurements concern detection, false alarms, and extrapolation-related release rather than the recovery rate of a search algorithm.

**Physics-informed discovery.** AI Feynman [7] uses physical units to constrain search; measurement-driven approaches recover governing structure from observations [8]; PySR [9] supplies a symbolic-search framework. Dimensional constraints and specification diagnostics provide different information. A dimensionally valid non-monomial law still misspecifies a power-law candidate. Our benchmark examines that form mismatch after fixing the candidate family.

**Position of the benchmark.** The study contributes a common experimental record linking selected diagnostic gates to mismatch directions and complete-workflow outcomes. Its comparative scope covers permutation curvature, fitted-value and regressor-power RESET, HC3 inference, and one nested-quadratic rival. Other residual tests, rival families, and post-selection procedures can be evaluated within the same accounting framework. The experiments establish findings for the tested gates and sampling designs; the wider specification testing literature remains the context for interpreting them.

# 3 Benchmark Design

## 3.1 Design principles and historical policies

**P1 - Explicit decision policies.** Each comparison identifies the statistic, threshold, eligible population, and handling of declined evaluations. Native p-value rules use alpha=.05. The permutation curvature p-value uses B=999 residual permutations; RESET uses an F reference. S5 compares native selection thresholds with H0-reference-calibrated thresholds. These policies are labeled separately in the results.

**P2 - A common reference target for headline power.** Table 3 calibrates each method to the empirical 5% quantile of its finite H0 p-values within the same sample-size stratum. Section 4.2 states the inclusive quantile convention and the resulting scope of the comparison. Native-size results, the mixed policies of S5, and joint panels retain their measured error rates. These are historical reference-pool results. Sections 3.8-3.9 define the separate, independently calibrated validation policies.

**P3 - Structure-defined labels.** H0 denotes a strict monomial whose mean is exactly linear in log-log coordinates; H1 denotes a non-monomial ground-truth structure. The symbolic registry supplies these labels rather than diagnostic outcomes. S3 family labels and S4 truth families likewise derive from their generators. Operational eligibility additionally requires valid data and the recorded consistency checks.

**P4 - Explicit failure registration.** Invalid evaluations retain a typed reason: zero-residual degeneracy, rank deficiency, insufficient degrees of freedom, leverage or covariance degeneracy, statistic mismatch, generation failure, or unavailable observational data. Each table states its evaluated denominator. A declined diagnostic cannot produce a rejection or contribute to joint release. Zero-noise cells exercise the degeneracy path.

**P5 - Source sampling configurations.** S2/S3 reuse the source registry's per-variable sampling objects. Tasks incompatible with the positive-input/output requirement are marked not applicable. The observational suite uses World Bank Open Data [14], with its indicators and cached snapshots recorded in the artifact.

## 3.2 Gates under test

All gates operate in log-log space on positive data: let ly = log y and X = [1, $lx_1$, …, $lx_k$] be the design of the candidate power-law model.

**A. Permutation curvature (form_check).** Adds a single mean-centered quadratic column q = $\text{mean}_j((lx_j - \bar{lx}_j)^2)$ and takes the added coefficient $c_2$ as statistic; size evidence comes from B = 999 residual permutations (vectorized through the pseudo-inverse row), with a statistic-mismatch sentinel comparing wrapper and native values on every call. Validity note: permuting *estimated* residuals assumes they are exchangeable, which OLS residuals are not exactly - the permutation null here is an approximation whose quality is an empirical question, not a finite-sample exactness theorem. The benchmark measures this construction's empirical size on the specified H0 pools, including deviations from .05 (Tables 1-2). Log-transformed multiplicative noise and favorable designs may matter, and §6 lists the stress tests (leverage-adjusted residuals, wild and block nulls, cluster dependence, fixed-design variants) that would bound where the approximation breaks.

**B. RESET fitted, classical F.** Adds $\hat{y}^2$ and $\hat{y}^3$; classical F test of joint significance.

**C. RESET fitted, HC3-Wald.** Same augmented design; HC3 sandwich Wald statistic on the added coefficients, referred to F(q, n − k).

**D. RESET exog, HC3-Wald.** Built regressors are powers of each log-regressor; HC3-Wald as in C.

**E. CV-quad (S5).** The "obvious alternative": rival = full quadratic expansion of the log features (linear terms, per-feature squares, pairwise cross-terms; minimum-norm least squares absorbs rank deficiency). Statistic D = RMSE_cv(power) − RMSE_cv(quad) over 5 deterministic folds; escalate if D > δ. Naive δ = 0; calibrated δ* chosen on the H0 pool to FPR = 5%.

**F. AICc-quad (S5).** Same rival; escalate if AICc(power) − AICc(quad) > m, k = column rank. Naive m = 0; calibrated m* as in E.

Historical A-F configurations were fixed for S1-S7 - powers (2, 3); B = 999; α = 0.05; the rival family and criteria of E/F; the grids below - and are identical across their applicable suites. Within each comparison the harness invokes its specified gates on the same datasets. The new validation evaluates A-G together; it does not merge datasets across historical suites or equate their different threshold policies.

## 3.3 Task suites S1-S3

**S1 - Synthetic control (19 scenarios × 500 seeds).** Three monomial H0 scenarios (univariate; two-variable; two-variable with ρ ≈ 0.9 correlated design). Eleven structural H1 scenarios: quadratic curvature (c = 0.10); cubic (standardized odd term); multiplicative interaction (log-space cross term); threshold (exponent 0.4 → 1.0 at the domain midpoint); exponential law; logarithmic law; rational (saturating) law; omitted variable; dilution (one of two variables curved); opposing curvature (+0.12, −0.12 on an exchangeable design, so the mean-square column's population coefficient is exactly zero); collinear-quadratic (ρ ≈ 0.95 design with curvature). Five noise-layer H1 scenarios have strict-monomial mean structure but non-Gaussian noise - Student-t(3); SD growing with the log prediction (0.3× to 1.7×); 2% point outliers (+8σ); 6% group outliers (+5σ); AR(1) ρ = 0.6 - rejections there are false alarms by construction and are never pooled into structural power. Grid: n ∈ {50, 200, 1000}; relative Gaussian noise σ ∈ {0, 0.5%, 1%, 3%, 5%} in log space; 500 seeds per cell via a seed rule embedding the grid position.

**S2 - SRSD-Feynman tasks (32 originally fitted tasks, 500 seeds).** The original positive-input screen selected 13 monomial H0 and 19 non-monomial H1 tasks from the 120-equation registry. Each fit has a 2,000-sample same-distribution test set. The extrapolation code constructs 12 log-spaced probe points along the first feature: six below the observed training minimum (down to minimum/3) and six above the observed maximum (up to 3 times maximum); all other log-features are held at their training medians. The cached far-field metric used in this paper is the relative error at the last, upper-end point only, against the noiseless closed-form truth. It is not a one-decade probe or the maximum error over the probe grid. Its coordinate depends on the observed training maximum and can change with n and seed. The generated fitted-run count is 240,000; downstream eligibility excludes i.30.5 after its sentinel failure and excludes i.26.2 only when a finite extrapolation outcome is required.

**S3 - Family-stratified Feynman equations (30 equations × 30 seeds).** Structural family classification from symbolic form: exact power law (12 H0); fractional power (1 H0); exponential (3 H1); rational (6 H1); trigonometric (5 H1); plus not-applicable members with explicit reasons. Each H1 equation runs over relative noise {0.5%, 1%, 3%, 5%} × {homoskedastic, heteroskedastic} × sampling-range scale {official, ×0.5, ×2}; H0 equations run the noise grid at official range.

## 3.4 Suite S4 - the family-viability battery

The battery evaluates four families through their linearizing coordinates: power (log x, log y), exponential (x, log y), logarithmic (log x, y), and rational (1/x, 1/y). Each member applies A's quadratic construction in its own coordinates and returns a viable set V={F: p_F ≥ alpha}. Columns are standardized before squaring. The formal run uses a wild-bootstrap null with Rademacher signs and B=999. Section 4.6 compares this choice with residual permutation.

The set-level outcomes are false exclusion of a true family, inclusion of a wrong family, and escalation when no family survives. Escalation is correct when the truth is outside all four families. The surviving set supplies candidates for further comparison rather than a unique selected model.

The grid has n=200, relative noise from 0.5% to 5%, 500 seeds per cell, and 188 cells (94,000 datasets). It includes the 19 S1 scenarios with generator-defined truth-family labels and 14 SRSD H1 equations under

homoskedastic and heteroskedastic noise.

## 3.5 Suite S5 - the gate comparison

S5 compares permutation curvature with the full-quadratic rivals E (cross-validation) and F (AICc), as specified in gate_compare.py. Its pools comprise the 19 S1 scenarios and the 32 originally fitted S2 tasks, with n in {50, 200}, rel=.03, and 200 seeds per cell. A, E, and F evaluate identical datasets.

A uses its native permutation rule. E/F are evaluated at both native selection thresholds and per-pool, per-n H0-reference thresholds. Tables 11-12 report measured size and detection under these policies. Because the resulting budgets differ, the comparison is interpreted using the reported H0 rates.

Reported outcomes include structural-H1 detection, noise-layer false alarms, task-level detail, and computational cost. The S7 reanalysis uses the same records to compare A alone with the A/E/F panel and reports the accompanying change in joint H0 rejection.

## 3.6 Suite S6 - real data (World Bank open macro data)

Synthetic and equation-derived suites have ground truth by construction; the deployment question is whether the synthetic failure-mode map predicts anything on observational data, where no truth exists. S6 is deliberately small, prespecified, and divergence-oriented: six macroeconomic relations (target, features) × four years (2019, 2021, 2022, 2023), cross-country cross-sections of 184-199 sovereign economies (World Bank aggregates excluded), from eight indicators pulled from the World Bank API with raw responses cached for reproducibility - life expectancy ~ GDP per capita (PPP); infant mortality ~ GDP per capita; GDP ~ population; $CO_2$ per capita ~ GDP per capita; internet penetration ~ GDP per capita; and life expectancy ~ GDP per capita + health expenditure per capita (two-feature). Gates A-D run at 5 permutation seeds per cell. These seeds are computational replicates of one p-value estimator on one fixed dataset - not five independent samples - so they are used only to report Monte Carlo stability (the empirical p has resolution 1/1000 at B = 999; across the 24 S6 cells the median p spread is .009 and the max .094), never as a sample-size or stability claim; a prespecified probe adds three agricultural and energy relations over three years, including a built-in control-contrast pair - cereal yield ~ fertilizer (one feature) versus cereal yield ~ fertilizer + energy use per capita (two features) - chosen to test, on live data, whether the dilution mechanism of S1 (a second correlated curved feature halving the mean-square column's signal) operates outside the laboratory. No calibration or power claim is made on S6; its contract is external validity of *mechanisms*, not of rates. An engineering companion bench (Appendix E) extends the same contract to physical-engineering data: nine open sources, 91 datasets, 150 prespecified relations, run as a single-realization smoke pass.

## 3.7 Suite S7 - joint-silence reanalysis of existing results

S7 evaluates panel non-rejection before imposing the fit screen. The A-D arm joins S2 records to SVD C/D outputs one-to-one on (task, n, rel, seed), requires positive noise and a valid statistic sentinel, and contains 18 H1 tasks and 13 H0 tasks. At each n in {50, 200, 1000}, a task contributes four noise levels times 500 seeds. Native alpha=.05 and per-gate H0-reference 5% thresholds define separate policies. Joint silence requires finite p-values for every gate and no rejection.

The A/E/F arm uses the S5 SRSD pool at rel=.03: 19 H1 tasks and 13 H0 tasks, with 200 seeds each at n=50 and 200. A retains its native rule; E/F use their individual H0-reference thresholds. A alone and A/E/F are compared on the same records, with the change in joint H0 rejection reported alongside detection.

The two arms have different task pools, noise levels, seeds, and gate sets. They estimate their own panel behavior; only the paired within-arm comparison measures the effect of adding E/F. Both arms reuse existing fitted records. Reference thresholds and the reported H0 rates use the same pools, so the evidence describes reference-pool calibration. audit_s7.py records filters and exact counts.

Section 4.10 additionally requires finite fit-quality and far-field outputs, yielding 17 H1 tasks. This defines the joint-release pool; S7's gate-only analysis uses all 18 eligible H1 tasks.

## 3.8 P0-1 v2 - independently calibrated joint release

P0-1 v2 separates reference fitting, threshold calibration, and testing. It uses the 32 historically evaluated equations (13 H0, 19 H1), n ∈ {50, 200}, and Gaussian log-noise SD 0.03. Twenty independent reference/calibration pairs contain 16,640 reference and 33,280 calibration datasets; one separate shared test bank contains 8,192 datasets (128 test replicates per equation and sample size). The two sample sizes have disjoint test datasets (4,096 each); every fitted policy is evaluated on the 4,096 datasets generated at its own n, so the reported denominators are 128 records per equation per sample size. All 58,112 datasets are newly generated. The historical equations are known tasks; new seeds alone do not establish structural generalization.

The first calibration pilot mapped statistics through a finite empirical H0 CDF. Distinct values beyond the reference maximum then shared an artificial ceiling, which could make a strict-budget panel unable to alarm. The frozen v2 protocol replaces this map. For A-D, u = -log10(max(p, float64.tiny)); for E/F/G, u is the raw rival statistic. A valid H0 reference pool supplies each method's median center and IQR/1.3489795 scale, falling back to normalized MAD and then population SD. An unusable scale or invalid member declines the policy. Valid standardized scores are not clipped to the reference range. A's finite 999-permutation resolution and floating-point p-value ties remain intrinsic limitations and are counted separately.

Five policies are prespecified: D; F; the A/B/C/D panel (AD); the A/E/F panel (AEF); and G. A panel uses the maximum of its member standardized scores. An independent H0 calibration pool fixes one threshold for the complete policy at withholding budgets 1%, 5%, and 10%. Withholding includes alarms and declines; strict score > threshold defines an alarm. If declines alone exceed the integer calibration budget, the policy is infeasible rather than silently treated as compliant. Release requires a training log-space $R^2 \geq 0.99$ screen, valid policy evaluation, and no alarm. Calibration uses the screened H0 population. The same target budget is not a guarantee of equal or bounded independent-test H0 withholding.

G is a fixed competitor: a 32-landmark RBF residual model, five-fold cross-validation, training-fold-only scaling, squared bandwidth equal to feature count, and ridge penalty 0.01 per observation. Its statistic is the power-law model's CV error improvement under this competitor. The kernel, complexity, and penalty are not tuned to H1 outcomes. G is a finite fixed-rival diagnostic; we make no omnibus-consistency claim.

The v2 outcome retains the sample-dependent legacy endpoint so that the calibration change is distinguishable from a domain change. The primary estimator is the equal mean of round-specific pooled-record ratios. We report calibration-round SD and empirical percentile ranges as sensitivity summaries, conditional paired task bootstrap intervals, round-only intervals, and intervals crossing rounds with H0/H1- stratified task weights (2,000 draws). All policies share resampling weights; undefined ratios remain undefined. These intervals condition on the known H0 calibration equation identities, realized calibration rounds, and the test bank. They neither redraw the calibration equation population nor include an inner record-level test bootstrap.

## 3.9 P0-2 - fixed external regions and structural-group holdout

P0-2 freezes a registry of 110 tasks before noisy training sampling: 32 historical tasks, six newly evaluated source equations with explicit positive-domain restrictions, 48 designed positive-mechanism tasks, and 24 new H0 designs. All historical source tasks and algebraic near-twins belong to development. H1 structural groups are assigned whole to development, confirmation, stress, or reserve. Global prefactors, renamed variables, nonlinear-core variants, and parameter windows cannot turn a known structure into an independent holdout. The source inventory is visible during protocol design, so confirmation means newly evaluated grouped tasks, not unseen formula metadata drawn from an unknown population.

The confirmation pool has 38 H1 tasks in eight structural groups and 12 H0 tasks in four design groups. Six of the H1 tasks derive from three physical-source groups: Gaussian profiles, a positive logarithmic source formula, and hyperbolic response. The remaining 32 tasks are controlled positive mechanisms, identified as such; they are not presented as additional physical laws. Each designed mechanism has a positive multiplicative amplitude input and four prespecified training-window variants. Its confirmation mechanisms include erf, atan, elliptic and Bessel functions, damped resonance, a nested exponential, softplus, and sech; the full formulas and bounds are in the frozen registry. Amplitude variation changes log-target variance and therefore $R^2$ screening; the resulting challenge pool is not a natural-frequency sample of physical mechanisms. Eight tasks in two reserve groups have no study samples, candidate fits, or gate outcomes. Deterministic software/domain checks do not count as experimental evaluation of those reserve groups.

Every H0 is a strict power law. Calibration uses 12 newly defined log-uniform H0 tasks; confirmation uses 12 distinct exponent designs sampled from Beta(2,2) in log coordinates. Groups correspond to the one-to-four-feature sampling designs. This is held-out design transfer within the monomial null family, not holdout of different null functional families. The fixed $R^2$ screen can itself discard correct low-signal models; conditional diagnostic cost and whole-workflow H0 retention are both reported.

For each task, the registry states train and outer bounds, coordinate scale (log, linear, or integer), registered domain restrictions, and an analytic validity argument. Positive inputs and positive finite truth are required throughout the complete registered outer region. Constants and dimensional source expressions are retained. Variables declared integer by the sampling schema use integer support; that schema is not a complete audit of every textual physical type. Stable expm1 evaluation replaces cancellation-prone exp(t)-1 without changing the mathematical source. For historical I.41.16, stable truth differs by at most 3.42% from the cancellation-prone upstream callback and agrees with a 65-digit calculation to $1.68\times10^{-15}$ relative error in preflight. This task remains development, and the old

experimental files are unchanged. No task, domain, window, or point is adjusted after observing fit quality, gates, or extrapolation error.

A deterministic external grid and its noiseless truth are saved before training draws. The recipe specifies eight coordinates per axial direction and 256 joint Sobol candidates plus outer corners, yielding 40-320 distinct external points after geometric filtering and merging. Axial external points receive half the mass; joint Sobol external points and outer corners receive half. Coordinates inside the registered training box are excluded by a fixed geometric rule. Duplicate coordinates merge with their weights summed. The same grid is used for every n, noise level, and training replicate. The primary endpoint is weighted mean pointwise relative error, sum w_j |exp(pred_log_j - log truth_j)-1|; dangerous H1 means this quantity >10%. Registered secondary endpoints are weighted p95 and maximum relative error; mean-error thresholds 2%, 5%, and 20% are sensitivity analyses. No denominator floor or finite-point deletion is used. Error targets the positive noiseless function (the log-noise location), not the lognormal arithmetic mean. These are finite-grid summaries, not continuum error bounds. A legacy endpoint is evaluated only when it lies inside registered valid support, and remains compatibility-only.

The full frozen grid is $n \in \{50, 200, 1000\}$ crossed with log-noise SD {0.005, 0.01, 0.03, 0.05}. Stage 1 covers n=50 and 200 at SD 0.03; stage 2 covers the other ten strata without rerunning stage 1. There are 100 independent test datasets per task and stratum. Ten independent reference/calibration pairs per stratum contain 32 reference and 64 calibration draws per calibration-null task. P0-1 v2's score transformation, policies, and budgets are reused; reference scales and thresholds are refitted only to these new H0 datasets. Formula registry, grouping, grids, seeds, and the complete noise/sample-size grid remain frozen across stages.

Five named evaluation pools separate confirmation, physical-source confirmation, designed-control confirmation, development, and stress. The physical and designed subpools share confirmation H0 controls and may share structural groups; they are not independent replications. The stress threshold mechanism includes departures absent from some training windows and does not determine the main policy ranking.

Primary points again average round-specific pooled-record ratios. The primary pool is confirmation-all at the registered 5% calibration budget. Resampling crosses the ten calibration rounds with H0-design/H1-structure groups (2,000 paired draws), keeping all variants in a group together and group weights shared across methods and selected rounds. Conditional-group and round-only intervals are also retained. An equal average of defined group-specific dangerous- release ratios is secondary; its omitted groups are counted. Zero risk denominators and undefined bootstrap draws are recorded, never converted to zero risk. Calibration percentile ranges describe ten-round sensitivity, not confidence coverage. These intervals remain conditional on the fixed calibration-design pool and realized test records; simulation volume cannot replace additional physical groups.

# 4 Results

Sections 4.1-4.11 retain historical S1-S7 results with their original policies. Sections 4.12-4.13 supply the new validation evidence. Tables 1-2 report size; Tables 3-4 detection; Table 5 the family map; Table 6 noise-layer behavior; Table 7 extrapolation; Tables 8-10 the battery; Tables 11-12 the model comparison; Tables 13-14 observational examples; Table 15 joint release; and Table 16 joint silence. Each table defines its event, denominator, task pool, and failure policy. Computational repeats are not independent equations. S7 reuses S2/S5 records rather than adding new fits. Sections 4.2, 4.10, and 4.11 apply the eligibility rules in Section 3 and the accompanying record-level audit scripts.

## 4.1 Size: false-positive rates on true monomials

**Table 1 - S1 H0 false-positive rates at p < .05. Three scenarios, four positive noise levels, and 500 seeds give 6,000 valid evaluations per n. Entries are rates with fit-level Wilson 95% intervals.**

| Method | n=50 | n=200 | n=1000 |
|---|---|---|---|
| A | .0583 [.0527-.0646] | .0457 [.0407-.0512] | .0513 [.0460-.0572] |
| B | .0557 [.0501-.0618] | .0518 [.0465-.0577] | .0472 [.0421-.0528] |
| C (SVD) | .0602 [.0544-.0665] | .0568 [.0513-.0630] | .0478 [.0427-.0535] |
| D (SVD) | .0675 [.0614-.0741] | .0525 [.0471-.0584] | .0502 [.0449-.0560] |

The evaluated-H0 denominator excludes 1,500 zero-noise records per n, each registered as degenerate_zero_residual. Such a record has no valid test decision and is not a successful non-rejection.

**Table 2 - S2 H0 false-positive rates at p < .05. Thirteen equations, four positive noise levels, and 500 seeds give 26,000 valid evaluations per n; C/D use SVD.**

| Method | n=50 | n=200 | n=1000 |
|---|---|---|---|
| A | .0489 [.0463-.0516] | .0535 [.0508-.0563] | .0517 [.0491-.0545] |
| B | .0499 [.0473-.0526] | .0497 [.0471-.0524] | .0527 [.0500-.0555] |
| C (SVD) | .0565 [.0538-.0594] | .0531 [.0505-.0559] | .0549 [.0522-.0577] |
| D (SVD) | .0604 [.0576-.0633] | .0564 [.0536-.0593] | .0583 [.0555-.0612] |

Both tables are generated from existing positive-noise H0 records by audit_size.py. Stored rejection flags agree with strict p < .05; ties at exactly .05 are not rejections. This differs from the inclusive empirical-quantile convention used in Table 3.

On synthetic Gaussian data, Table 1 shows small-sample size distortion: at n=50 all four fit-level Wilson intervals lie above .05, while the pattern is milder at larger n. Calibration is therefore empirical and imperfect, not exact. On the physics-derived designs (Table 2), A and B are near .05, while C and D have a modest upward size distortion: all six displayed C/D fit-level Wilson intervals exclude .05. The SVD calculation underlying these values was cross-checked against a separate implementation (Section 5.2).

Generalization over physical equations requires an equation-level uncertainty analysis, distinct from a fit-level interval on this fixed task pool. At n=200, resampling the 13 H0 equations gives C=.053 with a 95% interval [.050, .056] and D=.056 with [.055, .058]; the corresponding leave-one-equation-out ranges are [.052, .054] and [.056, .057]. The C lower bound rounds to .050 but is slightly above .05 before rounding. Appendix C uses this same SVD result and n. These intervals describe the observed task pool, not all possible physical laws.

A separate fitted-HC3 implementation check found 8/120 rejections (.067, Wilson [.034, .126]) on two tasks. This small-sample check addresses numerical agreement; Table 2 and the equation-level analysis supply the size estimates for the S2 pool. Appendix G records the original-path comparison and its interpretation.

## 4.2 Power at a common reference-pool calibration target

Table 3 uses the same 18 sentinel-valid H1 tasks for every method; feynman-i.30.5 is excluded. Each n has 36,000 records: four positive noise levels times 500 seeds per task. Thresholds are the empirical 5% quantiles of each method's finite H0-reference p-values at the same n. Table 3 rejects when a valid p-value is less than or equal to that quantile; native decisions instead use strict $p < .05$. Non-finite p-values and typed declines are never rejections or certificates. Quantile discretization and boundary ties mean that this is a calibration target, not a claim of exactly equal rejection counts or independent calibration.

**Table 3 - S2 rejection fraction on the common 18-H1-task pool at the H0-reference 5% target; C/D use SVD. Declines remain in the denominator and are not counted as rejections.**

| Method | n=50 | n=200 | n=1000 |
|---|---|---|---|
| A | .386 | .486 | .570 |
| B | .187 | .254 | .383 |
| C (SVD) | .128 | .221 | .383 |
| D (SVD) | **.466** | **.657** | **.718** |

For B and C, respectively, 2, 56, and 591 p-values are non-finite out of 36,000 at n=50, 200, and 1000. These records cannot contribute a rejection or a silent certificate. The displayed fractions are unconditional detection yields on the specified records, not power conditional on successful evaluation.

D has the largest mean rejection fraction among A-D in this pool. The paired equation-level bootstrap compares D minus A on these same 18 tasks (10,000 resamples): +.080 [-.083, +.256] at n=50, +.170 [+.042, +.329] at n=200, and +.148 [+.022, +.302] at n=1000. The first interval crosses zero; the latter two favor D in this task pool. D exceeds A on 10/18, 13/18, and 8/18 tasks, with 0, 1, and 5 ties, respectively. A larger pooled mean is therefore not uniform task-wise dominance.

Table 3, its paired effects, and Figures D.2-D.3 use the same eligibility rule. S5 evaluates the nested-rival gates on a separate pool; the paired D-versus-F validation is reported separately in Sections 4.12-4.13. The constructed alternatives below illustrate the method-specific differences behind the pooled ranking.

**Table 4 - S1 blind-spot cartography (raw $\alpha$, n = 200, rel = 1%).**

| Scenario | A | B | C | D |
|---|---|---|---|---|
| cubic (standardized odd term) | .40 | 1.00 | 1.00 | 1.00 |
| interaction ($lx_1 \cdot lx_2$ cross term) | .21 | 1.00 | 1.00 | .10 |
| opposing curvature (+c, −c, exchangeable) | **.02** | .98 | .98 | 1.00 |
| dilution (one curved variable of two) | 1.00 | .88 | .87 | 1.00 |

Three mechanisms, each verified by construction: (i) A's statistic is the mean of squared centered columns - under opposing curvature on an exchangeable design that mean has exactly zero population signal; (ii) a quadratic-projection statistic loses power on odd-symmetric deviations; (iii) B's fitted-value powers are a local Taylor surrogate for any smooth alternative - saturating on interaction where A (.21) and D (.10) struggle - and, as S3 shows, the same surrogate property near-blinds B on rational laws.

## 4.3 Family-stratified results and sensitivity

**Table 5 - S3 calibrated power by family (C/D on the verified SVD path).**

| Family | A | B | C (SVD) | D (SVD) |
|---|---|---|---|---|
| exponential | .09 | .06 | .07 | .06 |
| rational | **.65** | .16 | .13 | **.67** |
| trigonometric | .66 | .69 | .58 | **.86** |

B - near-perfect on synthetic quadratics - is nearly blind to rational laws (.16 calibrated): the fitted-value Taylor surrogate absorbs precisely the low-order curvature by which a saturating law departs from a power law. Under the SVD path D is the strongest gate on both rational (.67) and trigonometric (.86) families; the previous draft's near-zero C/D entries on trigonometric were the pinv artifact (thresholds derived from artifact p-mass), and the S3-level correction agrees in direction with the S2-level one of Table 3. Exponentials defeat every gate (≤.09 - the linearization a log transform induces on them is nearly orthogonal to all four augmentation families); A and D share the rational lead; no method dominates all three families. Every method's strength sits on another method's blind spot; Table 5 has no row-major winner. Sensitivity: scaling every sampling domain ×2 moves exactly one family-method cell by more than 15 points (rational × A: .66 → .84, the expected direction); ×0.5 moves none; heteroskedastic replications agree within .02 on every cell. **These sensitivity readings rest on 30 seeds per cell and cannot be separated from sampling noise at that size**: a single cell at p ≈ .66 carries a 95% interval of roughly ±.17, and the standard error of a difference between two independent 30-seed cells is ≈ .11, so a 15-point trigger cannot distinguish movement from variation. Family × method intervals are now attached (`paper/review3_analysis.py` §3.8b). On the within-cell binomial convention the trigonometric D (SVD) cell is .860 [.848, .871] and rational D is .669 [.655, .683]; on the honest unit - the equation - the same cells read .860 [.592, .998] and .669 [.409, .864]. Rational C moves from .132 [.122, .142] to [.053, .277] the same way. The equation-level intervals overlap across families and are wide enough to contain orderings the point estimates contradict, so the family ranking above is directional, not an ordered effect list; what survives the wider intervals is the sign pattern - D ≥ C on rational and trigonometric, every gate low on exponential - not the point ordering.

## 4.4 Noise-layer robustness

**Table 6 - S1 noise-layer H1 (strict-monomial mean structure) rejection rates at α = 0.05 - false alarms attributable to the noise assumption (n = 200, rel = 1%, 500 seeds; C/D on the verified SVD path).**

| Scenario | A | B | C (SVD) | D (SVD) |
|---|---|---|---|---|
| Student-t(3) noise | .034 [.021-.054] | .036 [.023-.056] | .030 [.018-.049] | .030 [.018-.049] |
| variance growing with prediction | .048 [.032-.070] | .058 [.041-.082] | .050 [.034-.073] | .050 [.034-.073] |
| AR(1) errors, ρ = 0.6 | .038 [.024-.059] | .040 [.026-.061] | .042 [.028-.063] | .042 [.028-.063] |
| 2% point outliers (+8σ) | .040 [.026-.061] | .066 [.047-.091] | .036 [.023-.056] | .036 [.023-.056] |
| 6% group outliers (+5σ) | .038 [.024-.059] | .060 [.042-.084] | .054 [.037-.077] | .054 [.037-.077] |

*Wilson 95% intervals use 500 seeds per cell. C/D are computed from the cached SVD results; paper/review3_analysis.py §3.8c documents the cell-level check.*

The recorded rates span .030-.066, with Wilson intervals shown (500 seeds per cell). These selected S1 perturbations do not isolate the causes of the S2 size distortions. Neither noise-related effects nor interactions between noise, design, and inference are ruled out.

## 4.5 Do alarms predict extrapolation failure?

The S2 extrapolation pool contains 180,000 records: 13 H0 and 17 H1 tasks, positive noise, and all three sample sizes. It excludes sentinel-invalid or missing-outcome tasks. Danger here means the registered upper-end point error exceeds 10%, without an $R^2$ screen; 36,206 records meet it (base rate .2011), including 37 records from true-form H0 tasks.

**Table 7 - Alarm precision and recall for >10% upper-end extrapolation error on the S2 pool, native thresholds; C/D use SVD. Declines produce no alarm and remain in the recall denominator.**

| Method | precision | recall | declines |
|---|---|---|---|
| A | .4226 | .6242 | 0 |
| B | .6379 | .4901 | 649 |
| C (SVD) | .6189 | .4639 | 649 |
| D (SVD) | .4879 | .8844 | 0 |

A and B have 10,296 and 5,442 exclusive true-positive alarms, respectively, with 12,304 shared. These counts show different alarm sets, not a claim of independence or chance-level overlap. D detects more of this pool's dangerous outcomes, but the precision/recall table does not compare methods at an equal false-positive budget. Figure D.4 uses the same eligible pool. An alarm and a non-rejection can both be informative under a specified sampling distribution; neither supplies a bound on error throughout an extrapolation domain.

## 4.6 The battery: from rejection to routing

**Table 8 - Probe: permutation vs wild null by family (n = 200, 200 seeds; entries rel = 1% / 3%).**

| Truth | perm null | wild null |
|---|---|---|
| power (size of power member) | .070 / .075 | .075 / .060 |
| rational: false escalation (V = Ø) | **.34 / .31** | .045 / .090 |
| rational: containment | .66 / .70 | **.96 / .91** |
| exponential: containment | .94 / .95 | .92 / .95 |
| logarithmic: containment | .87 / .93 | .89 / .95 |
| threshold (no battery family): escalation - *correct outcome* | 1.00 / 1.00 | 1.00 / 1.00 |

With the permutation null, the battery's own rational member rejects its own true family a third of the time - nothing about rational laws is at fault; the null is exchangeability-invalid in (1/x, 1/y) coordinates. The wild bootstrap repairs exactly that and nothing else.

**Set-level errors and multiplicity.** The battery's decision is the viable *set*, so the errors that matter are set-level, and each member runs at α = .05 without correction - a union decision whose two error types move oppositely under multiplicity. Two pools, two behaviors. On H0-side cells (truth is a battery member) the reported containment is one minus the exclusion rate of the true family's own member (.0496-.066 across the four truth families; the viable set is never empty *on this pool*), and per-member α does not inflate that rate because exclusion depends on one member only. Inflation appears on the other side: some wrong member survives alongside the truth in 23.3% of power-truth cells (near-neighbour overlap: exponential/logarithmic members retain power data) and 92% of logarithmic-truth cells (rational retains log data) - reported as near-neighbour overlap rather than suppressed in the locally recorded tables. On the structural-H1 pool (72,000 cells whose truth is no battery member by design) the viable set is empty - V = Ø, correct escalation - at the rates of Table 10's first column: universally for quadratic/threshold/omitted-variable (1.00), rarely for the interaction/opposing/collinear structures (.001-.05) whose mismatch directions the member families do not span. Routing there means *which wrong family is modal* (Table 10, second column); the battery is a candidate filtering stage whose surviving set the discovery layer must still rank, not a final selection. Downstream selection error after the battery is out of scope here and unmeasured.

**Table 9 - Battery on synthetic truths (500 seeds/cell, wild null).**

| Truth family | Containment (truth ∈ V) | False escalation (V = Ø) |
|---|---|---|
| power - size, 8 scenario groups incl. all noise layers | - | .0496 [.046-.053] pooled; per-scenario .043-.054 |
| exponential | .95 | .05 |
| logarithmic | .94 | .005 |
| rational | .93 | .066 pooled (≤ .082 worst rel) |

**Table 10 - Battery on out-of-family structures (escalation to V = Ø is the correct outcome; column 1 is the V = Ø rate).**

| Scenario | Correct escalation | Modal false V |
|---|---|---|
| quadratic / threshold / omitted variable | 1.00 / 1.00 / 1.00 | - |
| dilution | .92 | rational |
| cubic (odd term) | .24 | power (.68) |
| interaction | .05 | power (.95) |
| opposing curvature | **.00** | power+rational (.84) |
| collinear-quadratic | .03 | rational (.61) |

Table 10 is Table 4 re-expressed as routing errors: the power member inherits A's documented blind spots, so those structures stay in V under the power label instead of escalating. On the SRSD side the routing is reported descriptively (median |V| 1-2 for 11 of 14 equations; i.30.5 → {exponential} in 78%; ii.38.14 escalates on 67%), because surface-form family labels dissolve under examination - the battery correctly rejects Planck-type "exponentials" that no linearizable family owns.

## 4.7 The gate comparison: why not just compare against a nested rival?

**Table 11 - S5 size on H0 pools (200 seeds/cell; naive and calibrated thresholds for E/F).**

| H0 pool | A | E naive | E cal. | F naive | F cal. |
|---|---|---|---|---|---|
| synth n=50 (600 fits) | .080 | .212 | .050 | .132 | .050 |
| synth n=200 (600) | .040 | .183 | .050 | .135 | .050 |
| SRSD n=50 (2,600) | .051 | .167 | .050 | .086 | .050 |
| SRSD n=200 (2,600) | .038 | .152 | .050 | .123 | .050 |

**Table 12 - S5 structural-H1 detection (A at its native threshold; E/F at H0-pool-calibrated thresholds; mean over tasks).**

| | synth n=50 | synth n=200 | SRSD n=50 | SRSD n=200 |
|---|---|---|---|---|
| A (permutation curvature) | .774 | .782 | .348 | .493 |
| E (CV-quad) | .929 | .934 | .462 | .628 |
| F (AICc-quad) | **.930** | **.940** | **.525** | **.669** |

*OR(A, E) at each gate's own 5% threshold - a different, larger joint error budget (.065-.102 across the four benches), reported separately: .937/.941/.547/.675. It is not comparable to the rows above and is excluded from this table.*

The error budgets are not exactly matched in Table 12: A retains its native threshold, whereas E/F are tuned on the H0 reference pool. In particular, A's synth n=50 size is .080, and its SRSD n=200 size is below .05. Interval overlap with a target is not proof of an identical budget. Table 12 reports measured

detection under these policies, not a uniformly matched-FPR ranking.

The quadratic rivals include per-feature squares and cross-terms, covering directions absent from A's single mean-square column. On the recorded S5 pool, F has the largest mean detection in each displayed column. This is a scoped finding for one rival family, not a universal default recommendation.

Naive E/F thresholds have elevated measured H0 rejection rates (Table 11). Reference-pool calibration reduces them to the reported targets; A requires no separate reference pool to construct its resampling p-value, but that does not make A size-valid on every pool. Its .080 small-sample result is one counterexample.

The earlier SRSD leave-one-H0-equation-out analysis reports held-out sizes .053/.050 for E at n=50/200 and .051/.052 for F, with per-equation spread [.010, .095]. This provides a limited transfer check across the 13 available H0 tasks, not calibration on all new equations or a replacement for an independent deployment test set.

For OR(A,E), using the individual thresholds increases the joint H0 rejection budget to .065-.102. At synth n=50, A alone already exceeds .05, so tightening E cannot achieve a 5% union budget while preserving A's rule. The reported joint-budget recalibration at n=200 gives .940/.664 detection on the synthetic/SRSD benches, compared with F's .940/.669. Those specific comparisons do not establish that F is the frontier at every possible error budget. Combining gates and selecting thresholds should be evaluated as a complete decision policy.

## 4.8 Real data: does the map predict live behavior?

**Table 13 - S6 World Bank relations: verdicts (median permutation p over 5 seeds for A; deterministic p for B-D; α = .05).**

| Relation (target ~ features) | Years all four gates agree | Divergent cells (which gate, which way) |
|---|---|---|
| infant mortality ~ GDP pc | 4/4 reject | - |
| $CO_2$ pc ~ GDP pc | 4/4 reject | - |
| internet penetration ~ GDP pc | 4/4 reject ($p < 10^{-16}$) | - |
| GDP ~ population | 4/4 **quiet** (A .71-.91; B .19-.33) | - |
| life expectancy ~ GDP pc | 0/4 | A alone rejects 2019 (.035) & 2022 (.008); B alone rejects 2022 (.0002); C/D never |
| life expectancy ~ GDP pc + health exp pc | 2/4 quiet | B rejects 2022 (.0002) while A quiet (.108) |

Where signal is strong the gates agree - three relations rejected unanimously every year, one candidate relation unrejected by all four gates - and A's five p-values per cell are Monte Carlo replicates of one estimator whose spread is small (median .009, max .094 at resolution 1/1000; no verdict flips beyond boundary cells). Divergence concentrates where the synthetic map puts weak curvature: the life-expectancy cells flip year to year (signal at the detection boundary), and the two-feature cell shows B rejecting where A stays quiet. We label this pattern *consistent with* the dilution mechanism (§4.2, Table 4) rather than a reproduction of it: on observational data the p-value change after adding a second feature can

also reflect feature correlation, omitted-variable changes, sample composition, measurement error, cross-country heterogeneity, or spatial dependence - no ground truth exists to separate these, so the claim is signature-consistency, not mechanism confirmation.

**Table 14 - S6 dilution probe: the same target with and without a second correlated curved feature (3 years × 5 seeds; detection = median gate verdict).**

| Relation | A | B | D |
|---|---|---|---|
| cereal yield ~ fertilizer (1 feature) | rejects 3/3 years (.011-.031) | rejects 3/3 (~.003) | rejects 3/3 |
| cereal yield ~ fertilizer + energy pc (2 features) | **quiet 3/3 years** (.067-.149) | rejects 2/3 (.025-.092) | rejects 3/3 (≤.018) |

This matches the S1 DILUTION scenario's signature - adding one correlated curved feature silences A on a relation it was rejecting, while the RESET variants continue to alarm - a pattern *consistent with* the dilution mechanism rather than a demonstrated reproduction of it, since feature correlation, sample composition, and omitted-variable changes move the same p-values in the same direction. No ground truth exists on S6 and none is claimed; the contract of §3.6 is signature consistency in both directions: unanimous cells where the map predicts signal, and map-consistent divergence where it predicts weakness.

A companion engineering bench (Appendix E) pushes the same probe across nine physical-engineering sources (147 run cells): unanimous detection of a documented rating-curve misspecification, mechanism-echo divergences matching the synthetic map (small-sample power, interaction carried by a second feature, all three HC3 failure modes live), and mutual misses on locally approximate laws - all recorded, none adjudicated.

## 4.9 Runtime

The historical S1 runs recorded per-dataset means of 12.0-25.3 ms for A and 1.0-1.8 ms for B/C/D; the family battery at n=200 recorded 24 ms, and S5 E/F 1.5-3.5 ms. These figures describe the recorded environments and implementations. No inference-path runtime equivalence or production-throughput guarantee is claimed. Figure D.5 is retained as a historical timing panel; the SVD revision requires a separately labeled timing comparison.

## 4.10 Historical joint release at a sample-dependent endpoint

The joint-release analysis links fit quality, native A/B decisions, SVD C/D decisions, and extrapolation errors one-to-one on (task, n, rel, seed). It requires positive noise, a valid statistic-consistency sentinel, and finite $R^2$ and far-field outcomes. The 17 eligible H1 tasks contribute 34,000 records per n. The i.30.5 statistic-sentinel records and the i.26.2 records without finite far-field outcomes are excluded under these rules. audit_results.py records the join and all filters.

**Decision events.** Release requires $R^2 \geq 0.99$, four valid gate evaluations, and no rejection at the native alpha=.05 rules. A dangerous wrong model has relative error >10% at the registered upper-end probe (Section 3.3). A declined diagnostic withholds release. The error threshold defines the consequence

evaluated here; the probe measures one input rather than an operating-domain maximum.

**Table 15 - Joint release on the 17-task S2 H1 pool. Each n has 34,000 records. Count symbols S, Sd, Ld, and Da distinguish the event populations from gate labels A-D. Eligibility excludes i.30.5 (statistic-sentinel failure) and i.26.2 (no finite far-field outcome); Table 16 uses the 18-task H1 pool without the finite-far requirement.**

| n | Screened S | Dangerous in S: Sd | Released dangerous Ld | All dangerous Da | Ld / S | Ld / Sd | Ld / Da |
|---|---|---|---|---|---|---|---|
| 50 | 22,060 | 2,208 | 350 | 11,922 | 1.59% | 15.85% | 2.94% |
| 200 | 21,932 | 2,150 | 106 | 12,202 | 0.48% | 4.93% | 0.87% |
| 1000 | 22,000 | 2,045 | 25 | 12,045 | 0.11% | 1.22% | 0.21% |

S counts screened wrong models; Sd counts dangerous wrong models within S; Ld counts those passing both the screen and all four valid gates; Da counts every dangerous model in the eligible H1 pool. The three ratios answer distinct operational questions: frequency among screened wrong forms, failure to intercept screened-dangerous forms, and survival among all dangerous fits. Pooled across n, S=65,992, Sd=6,403, Ld=481, and Da=36,169. The respective rates are 0.73%, 7.5%, and 1.33%.

**The fit screen changes the release population.** At n=50, 966 dangerous records have four valid non-rejecting gates; 616 fail the fit screen, leaving 350 complete-workflow releases. At n=200 and 1000, the gate-only and complete-workflow dangerous release numerators coincide at 106 and 25. Across n, the panel declines on 2, 56, and 591 eligible H1 records. This accounting separates rejection, diagnostic withholding, fit screening, and complete release.

**Detection benefit and residual failure.** Among screened-dangerous wrong models, 481/6,403 (7.51%) are released. Among screened wrong models below the danger threshold, 27,025/59,589 (45.35%) are released. The panel therefore excludes most screened-dangerous records from release while retaining a substantial residual channel. These pooled rates summarize the sampled task composition, and that composition is concentrated. The 481 releases come from two of the seventeen tasks: bonus.15 (242) and bonus.5 (239). Six of the remaining fifteen release no dangerous record at any n while contributing 30,351 dangerous records between them; among these, all 557 dangerous records of ii.38.14 are intercepted. Structural mismatch and failure at the chosen probe are different labels, and their relation varies by equation.

**Sensitivity to the danger threshold.** At n=50, thresholds of 2%, 10%, and 20% label 34.91%, 10.01%, and 2.76% of screened H1 models dangerous. The release fractions within those subsets are 41.30%, 15.85%, and 6.40%. At n=1000 and the 20% threshold, Sd=0 and Ld/Sd is undefined. The complete 2/5/10/20/50% sweep is in evidence/results.json.

A pooled task-level sweep makes the composition visible. At the 10% threshold, all 481 releases come from bonus.15 (242) and bonus.5 (239). At 12.5%, the count falls to 176: bonus.15 contributes 5 and bonus.5 contributes 171. At 15%, all 111 releases come from bonus.5; at 20%, all 39 do. The released set therefore consists of the two tasks throughout, but which task dominates changes sharply between 10% and 12.5%. These counts are reproduced by threshold_decomposition.py and recorded in evidence/threshold_decomposition.json.

**Sample-size pattern.** Ld/Da decreases from 2.94% to 0.87% to 0.21%, and Ld/Sd decreases from 15.85% to 4.93% to 1.22%. A fixed-label control assigns each matched (task, rel, seed) triple its n=1000 danger label. Across 34,000 complete triples, Ld/Sd is then 16.77%, 1.87%, and 1.22%, while Ld/Da is 2.93%, 0.31%, and 0.21%. The decreasing pattern survives this label control. Its interpretation still combines fit-screen behavior, diagnostic sensitivity, and changing probe coordinates; training samples are not constructed as nested sets.

**Task composition of the released set.** The released records change identity with n. At n=50 the 350 releases are 239 from bonus.5 and 111 from bonus.15. At n=200 and n=1000 every release (106 and 25) comes from bonus.15 alone. Excluding that task, Ld/Sd is 11.58%, 0%, and 0%. The fixed-label control holds the danger label but not the task composition: bonus.15 has a median relative far-field error of 9.2%, with the 75th and 90th percentiles at 9.5% and 9.8% against the 10% threshold. Its danger label therefore moves with the threshold while its fitted form and task identity do not change. The decreasing pattern combines a panel effect with the position of one task's error distribution relative to a fixed cutoff.

**Task-level uncertainty.** The record-level ratios treat simulation replicates as independent observations. A task bootstrap instead resamples the 17 eligible equations with replacement (B=10,000), carrying each task's S/Sd/Ld/Da and recomputing the pooled ratios. Pooled over n, Ld/Sd is 7.5% with a 95% interval of [0%, 49.9%]; 12% of resamples contain no released dangerous record, and 15 of the 17 tasks contribute Ld=0. Leave-one-task-out values span [3.9%, 20.1%]: omitting bonus.5 raises Ld/Sd to 20.1%, while omitting bonus.15 lowers it to 3.9%, because the remaining task's own release fraction then dominates. At n=200 and 1000 the interval reaches [0%, 75.7%] and [0%, 56.8%], since only bonus.15 contributes releases. This interval measures between-equation heterogeneity, not binomial precision, and the record-level 7.5% describes the sampled task pool rather than a task-uniform rate. Counts and intervals are reproduced by audit_task_intervals.py and recorded in evidence/task_intervals.json.

**Information in a clean diagnostic record.** The corresponding 180,000-record H0/H1 pool has three states: true H0 (78,000 records; 65,934 releases), dangerous wrong H1 (36,169; 481 releases), and non-dangerous wrong H1 (65,831; 27,025 releases). The empirical release probability is .013299 for dangerous H1 and .845308 for true H0. Their likelihood ratio is .01573 in the dangerous-versus-true direction, or 63.56 in the reverse direction.

Across all three states, 481 of the 93,440 released models are dangerous wrong forms (0.515%). This proportion depends on the pool's class composition. In a different, explicitly hypothetical population containing only true H0 and dangerous H1 with equal priors, Bayes' rule gives a 1.55% dangerous posterior after release. These are separate populations and their conditional risks must remain separate. Applying the decision rule to a new population requires its own validation.

The results support a complete-workflow evaluation: diagnostics carry information and remove dangerous candidates, while a clean record leaves measurable residual uncertainty.

## 4.11 Historical joint silence without the fit screen

Panel non-rejection is measured separately from complete release using the S7 arms defined in Section 3.7. audit_s7.py reproduces these counts from the existing records.

**Table 16 - A-D joint silence on S2 records. H1 has 18 eligible tasks (36,000 records per n); H0 has 13 tasks (26,000 per n). Calibrated means a 5% H0-reference target per gate.**

| n | H1 native alpha=.05 | H1 per-gate calibrated | H0 per-gate calibrated |
|---|---|---|---|
| 50 | 13,616 / 36,000 (37.8%) | 13,888 / 36,000 (38.6%) | 22,821 / 26,000 (87.8%) |
| 200 | 10,717 / 36,000 (29.8%) | 10,877 / 36,000 (30.2%) | 23,102 / 26,000 (88.9%) |
| 1000 | 8,355 / 36,000 (23.2%) | 8,478 / 36,000 (23.6%) | 23,108 / 26,000 (88.9%) |

A silent record has finite p-values for all four gates and no rejection. The H0 rates are measured on the reference pool used to set the thresholds. The resulting joint H0 rejection exceeds 5%, illustrating the distinction between individual calibration targets and the error budget of a panel.

**Paired A versus A/E/F.** On the S5 SRSD pool at n=50, adding E/F reduces H1 silence from 2,479/3,800 (65.2%) to 1,602/3,800 (42.2%). H0 silence decreases from 2,468/2,600 (94.9%) to 2,352/2,600 (90.5%). At n=200, H1 silence decreases from 1,926 to 1,160 out of 3,800, and H0 silence from 2,500 to 2,377 out of 2,600. The paired comparison demonstrates increased detection together with increased false rejection. It motivates the equal-joint-budget validation in Sections 4.12-4.13.

This three-gate arm has 19 H1 tasks at one noise level; the A-D arm has 18 H1 tasks at four noise levels. The S5 cache has no recorded failures for i.30.5 but lacks S2's separate replay-sentinel fields. Its within-arm counts are retained as exploratory evidence. Comparison of complete A-D and A-F panels requires common records.

**Class-conditional interpretation.** Within the S5 A/E/F arm at n=50, silence favors true over wrong forms by the likelihood ratio (2352/2600)/(1602/3800)=2.146. Under an explicitly hypothetical 50/50 class prior, the posterior probability of a wrong form is (1602/3800)/[(1602/3800)+(2352/2600)]=31.8%. The likelihood ratio measures evidence under this policy; the posterior also uses the assumed prior.

Table 15 additionally imposes $R^2 \geq 0.99$ and requires a finite far-field outcome. At n=50, 22,060/34,000 eligible H1 records pass that screen (64.9%). Gate-only silence and screened dangerous release therefore answer different questions with their own eligible populations.

## 4.12 Fresh-data joint-policy validation and calibration uncertainty

All 600 P0-1 v2 policy configurations are feasible, giving 2,457,600 decisions on a shared independent test bank. AEF at n=50 and the 1% target has nonzero H1 alarms in all 20 rounds. This verifies removal of the artificial reference-CDF ceiling. It does not remove statistic ties from A's finite permutation resolution or floating-point p-values.

**Table 17 - P0-1 v2 at the registered 5% calibration budget. H0 withholding is alarm or decline conditional on $R^2$ screening. Dangerous release is conditional on screened H1 with legacy endpoint error >10%. Points average 20 calibration-round ratios. Brackets are crossed task/round 95% percentile intervals.**

| n | Policy | Test H0 withheld | Dangerous H1 released | Crossed interval |
|---|---|---|---|---|
| 50 | D | 4.12% | 34.72% | [0.00%, 56.42%] |
| 50 | F | 4.68% | 7.98% | [0.00%, 48.15%] |
| 50 | AD | 4.40% | 21.35% | [0.00%, 48.15%] |
| 50 | AEF | 4.67% | 7.98% | [0.00%, 48.15%] |
| 50 | G | 5.46% | 7.98% | [0.00%, 44.60%] |
| 200 | D | 5.06% | 5.56% | [0.00%, 87.50%] |
| 200 | F | 4.79% | 5.56% | [0.00%, 100.00%] |
| 200 | AD | 4.69% | 4.41% | [0.00%, 77.81%] |
| 200 | AEF | 4.22% | 5.42% | [0.00%, 95.62%] |
| 200 | G | 5.36% | 4.83% | [0.00%, 86.25%] |

The independent-test costs are measured outcomes, despite common calibration targets. The known-task evidence is too narrow for a general ranking. Screened-dangerous denominators come from six tasks at n=50 and four at n=200; AEF's dangerous releases still all come from bonus.15. At n=50 F has 7.98% dangerous release in every round. Its round-only interval is correspondingly degenerate, whereas its crossed interval extends to 48.15%. Stable recalibration therefore does not establish adequate equation coverage. At n=200, 19/2,000 risk bootstrap draws have no denominator and are recorded as undefined. All v1 files and v2 outputs remain separately preserved.

## 4.13 Fixed-region validation across the complete registered grid

Stage 2 completes the ten strata omitted from stage 1, adding 205,200 independent datasets in 3,300 execution cells. The full frozen n/noise grid therefore contains 246,240 datasets: 46,080 reference, 92,160 calibration, and 108,000 test. The primary confirmation pool comprises 60,000 distinct test cases (14,400 H0, 45,600 H1); development and stress account for the other 48,000. No stage-1 dataset, formula, domain, weight, endpoint, or policy definition is replaced. The eight reserve tasks remain without study sampling, fitting, or gate outcomes.

All 1,800 calibrated policy configurations are feasible. Their 16,200,000 decisions are stored in separate stage artifacts; repeated decisions on one shared test bank are not additional independent datasets. Independent stage audits reproduce fixed-grid error from saved coefficients, H0 reference scales and joint thresholds, all decisions, group event counts, and primary/equal-group/paired uncertainty. A separate complete-grid audit confirms nonoverlapping data streams, exact stratum/table integration, and preservation of the prior artifacts. These checks use recorded A-G statistics; they do not independently rerun every statistic from the original training observations.

**Table 18 - Complete P0-2 confirmation grid at the same 5% joint calibration budget. Values are mean dangerous H1 release conditional on $R^2 \geq 0.99$ and fixed-region weighted mean error >10%. Ten calibration rounds share the same independent test bank in each stratum. G is the frozen RBF rival. The last column counts H1 structure groups contributing to this primary danger denominator.**

| n | Log-noise SD | D | F | AD | AEF | G | H1 groups |
|---|---|---|---|---|---|---|---|
| 50 | 0.5% | 10.04% | 7.47% | 12.54% | 17.54% | 11.73% | 6 |
| 50 | 1% | 19.91% | 15.96% | 23.14% | 32.62% | 22.68% | 6 |
| 50 | 3% | 50.35% | 41.59% | 55.75% | 64.41% | 53.67% | 7 |
| 50 | 5% | 72.39% | 53.99% | 76.45% | 79.92% | 74.09% | 7 |
| 200 | 0.5% | 0.55% | 0.39% | 0.89% | 2.55% | 1.30% | 6 |
| 200 | 1% | 8.40% | 6.33% | 10.89% | 14.25% | 10.22% | 6 |
| 200 | 3% | 26.51% | 21.88% | 31.51% | 32.99% | 31.12% | 7 |
| 200 | 5% | 43.79% | 35.30% | 49.66% | 51.93% | 48.26% | 7 |
| 1000 | 0.5% | 0.00% | 0.00% | 0.00% | 0.00% | 0.00% | 6 |
| 1000 | 1% | 0.09% | 0.00% | 0.35% | 0.79% | 0.56% | 6 |
| 1000 | 3% | 11.45% | 9.36% | 13.59% | 15.65% | 13.30% | 6 |
| 1000 | 5% | 22.02% | 17.75% | 26.18% | 25.79% | 25.81% | 6 |

At the 3% log-noise level, F's mean dangerous release decreases from 41.59% at n=50 to 21.88% at n=200 and 9.36% at n=1000. AEF gives 64.41%, 32.99%, and 15.65%, respectively. At 5% noise, F still releases 17.75% of screened-dangerous H1 at n=1000, compared with 25.79% for AEF. All five policies have zero observed primary dangerous release in the n=1000, 0.5%-noise stratum. Across the complete grid, F has the lowest point in eleven strata, and all methods tie at zero in the remaining stratum. This describes this registered challenge library and budget; it is not a universal ranking or a test-cost-matched dominance claim. Figure D.6 displays the full-grid estimates with uncertainty.

The paired AEF-minus-F dangerous-release differences at 3% noise are +22.82 [17.93, 29.64], +11.11 [8.67, 13.17], and +6.29 [2.47, 9.00] percentage points for n=50, 200, and 1000. Corresponding D-minus-F differences are +8.76 [5.23, 12.15], +4.63 [3.35, 5.54], and +2.08 [0.75, 3.24] points. These are crossed group/calibration-round 95% percentile intervals; all 2,000 draws are defined for these comparisons. At n=1000 and 3% noise, F's individual interval is [3.54%, 13.76%] and AEF's is [6.27%, 22.23%]. Intervals are pointwise rather than simultaneous guarantees across strata. They remain conditional on the fixed calibration-design pool and realized test records. The limited conclusion is that adding panel members does not guarantee lower dangerous release at a common calibration budget on these controlled mechanisms.

Each noise level shows decreasing release points as n increases, but different n values use independent training samples and can have different screened-dangerous composition. The point pattern is not a nested-sample causal estimate of sample-size benefit. Six or seven H1 groups contribute primary danger in each stratum; all primary screened-dangerous cases come from designed mechanisms. Their distinct parameter windows remain grouped for uncertainty.

**Table 19 - Correct-model cost and physical-source denominator coverage for the same grid. H0 columns use the confirmation null designs. F/AEF withholding is conditional on the fit screen; F retention uses all H0 test cases. Physical Sd counts screened-dangerous physical-source H1 cases: the primary mean-error criterion and the registered p95 secondary criterion, both >10%. These are event counts, not physical risk rates.**

| n | Log-noise SD | H0 screen | F withheld | AEF withheld | F retained | Physical Sd | p95 Sd |
|---|---|---|---|---|---|---|---|
| 50 | 0.5% | 100.00% | 4.47% | 4.51% | 95.53% | 0 | 100 |
| 50 | 1% | 97.67% | 3.46% | 5.21% | 94.28% | 0 | 100 |
| 50 | 3% | 70.67% | 5.24% | 4.80% | 66.97% | 0 | 100 |
| 50 | 5% | 49.92% | 3.52% | 3.67% | 48.16% | 0 | 102 |
| 200 | 0.5% | 100.00% | 5.03% | 4.46% | 94.97% | 0 | 100 |
| 200 | 1% | 99.33% | 4.77% | 4.98% | 94.59% | 0 | 100 |
| 200 | 3% | 69.58% | 3.95% | 5.58% | 66.83% | 0 | 100 |
| 200 | 5% | 48.08% | 3.80% | 6.00% | 46.26% | 0 | 100 |
| 1000 | 0.5% | 100.00% | 7.13% | 5.42% | 92.87% | 0 | 100 |
| 1000 | 1% | 99.92% | 4.71% | 5.05% | 95.21% | 0 | 100 |
| 1000 | 3% | 68.67% | 4.25% | 3.25% | 65.75% | 0 | 100 |
| 1000 | 5% | 49.50% | 4.71% | 7.76% | 47.17% | 0 | 100 |

Correct-model loss is substantial at larger noise, even before a gate acts. At 3% noise the H0 screen passes 70.67%, 69.58%, and 68.67% for n=50, 200, and 1000; F's whole-workflow retention is 66.97%, 66.83%, and 65.75%. At 5% noise the screen passes only 48.08%-49.92% across n. Conditional H0 withholding also varies despite identical calibration targets: F ranges from 3.46% to 7.13% and AEF from 3.25% to 7.76%. In the low-noise n=1000 stratum, both release no observed primary dangerous H1 but F's test H0 cost is 7.13% versus AEF's 5.42%. The measured costs are integral to the comparison. The 5% calibration budget does not imply 95% overall retention or an independent-test size guarantee. Figure D.7 separates this loss from physical coverage.

The physical-source pool's primary conditional rate remains undefined in all twelve strata because its screened-dangerous denominator is zero. The complete workflow nevertheless releases 0 of 3,600 observed primary dangerous physical-source fits across the grid, an overall secondary result: all three Gaussian tasks' 1,200 fits apiece are dangerous and all fail the $R^2$ screen. The other three source tasks' 3,600 fits all clear the screen and none meets the primary danger criterion. Thus the fit filter intercepts the observed dangerous cases; this pool supplies no evidence on how the gates handle high-fitting but externally dangerous physical candidates. It is neither a post-screen zero-risk estimate nor a physical-population safety statement.

The registered physical p95 secondary criterion has 100 screened-dangerous cases in eleven strata, all from I.44.4. At n=50 and 5% noise it has 102, including two II.35.21 cases from a second group. At 3% noise, F's p95-dangerous release is 97.0%, 19.2%, and 0% across n=50, 200, and 1000; AEF's is 95.4%, 47.3%, and 0%. These are gate/workflow consequences for a secondary endpoint supported by one source group at that noise level. They do not promote that endpoint to the primary criterion or establish broad physical performance. At n=1000 the p95 intervals degenerate to zero on the fixed test bank, and only

1,393/2,000 crossed resamples have a defined denominator. Undefined draws remain undefined. Such single-group intervals do not acquire physical-population precision from the number of simulation replicates.

All data-generation, candidate-fit, and fixed-grid outcomes are valid. B and C each decline as rank_deficient on the same 513 development-test datasets; A, D, E, F, and G have no recorded failures. These are 513 affected datasets, not 1,026 independent failures. The declines occur in II.11.28 and III.12.43 and do not affect the confirmation pool. They remain in the registered development decision accounting, including AD withholding; none is deleted or regenerated. Full operational, threshold, endpoint, and danger-cutoff tables are retained in the stage and complete-grid artifacts.

# 5 Discussion

## 5.1 Match diagnostics to the mismatch direction

The controlled alternatives explain why a single method ranking is insufficient. A's mean-square column can cancel opposing curvature and has weak signal for odd-symmetric departures; interactions distinguish fitted-value augmentation from regressor-power augmentation (Table 4). Resampling calibrates a chosen statistic but does not expand the directions it examines. The task-level differences therefore matter alongside the average detection rates.

D is the strongest mean performer among A-D on the common S2 pool under the stated reference target. Its paired advantage over A is supported at n=200 and 1000. S5 separately shows that a richer quadratic rival has higher recorded mean detection than A in each displayed pool. Together these findings motivate testing both augmentation geometry and model-comparison alternatives. The new paired common-budget comparison in Section 4.13 evaluates D and F on the same records, while preserving its task-library and test-cost qualifications.

## 5.2 Separate statistical assumptions from numerical stability

Native size is an empirical requirement: Tables 1-2 show departures from .05 even when the symbolic mean is a true monomial. The family-viability experiment supplies a second example. In rational coordinates, changing the resampling null substantially improves observed containment (Table 8). Transforming coordinates therefore requires reevaluating the noise and resampling assumptions.

Numerical implementation supplies a distinct failure mode. A stratified cross-check of exog-HC3 on 176 fits spanning all 13 H0 equations compared the original pinv(X′X) path, an SVD path, and statsmodels on identical data. The SVD and statsmodels calculations agreed closely (maximum absolute p-value difference .0023), whereas the original path produced large rejection rates on near-collinear designs. Forming X′X amplifies conditioning problems; the stable path changes the empirical method comparison. The audit supports using the SVD results reported here. The complete measurements and earlier interpretation are preserved in Appendix G.

## 5.3 Measure useful rejection, correct-model loss, and residual release

The historical endpoint analysis establishes an observable residual channel: 481 of 6,403 screened-dangerous H1 fits pass its native four-gate policy. It also shows that those releases concentrate in two equations. P0-1 v2 makes independent joint calibration explicit, and P0-2 fixes the external coordinates and structural-group units. These layers answer related questions with different pools and outcomes; their percentages cannot be pooled into a single risk rate.

Adding members changes a panel's maximum-score null distribution and its jointly calibrated threshold. Consequently, extra members can reduce sensitivity to some departures even though a panel at unchanged member thresholds would alarm more often. The empirical comparison must therefore include the target budget, realized independent-test H0 withholding, and the H1 consequence on the same fits. A lower dangerous-release point at one budget does not establish dominance at identical test cost or on an arbitrary new population.

The $R^2$ screen deserves the same scrutiny as the diagnostics. The calibration budget concerns diagnostic withholding among screened H0 candidates; it does not guarantee 95% retention of all correct models. Whole-workflow H0 retention, fit-screen passage, alarms, declines, and dangerous release are separate outputs. An undefined screened-dangerous denominator is also a result about identifiability: it neither estimates the post-screen gate risk nor establishes interception by those gates. Overall dangerous release can still be measured against the pre-screen dangerous population, as in the physical-source results in Section 4.13.

Region-level error connects the decision rule to intended use more directly than one moving endpoint. The registered weighted mean, p95, and maximum still summarize a finite grid, use a specified relative-error tolerance, and depend on the registered domain. No diagnostic record supplies a distribution-free extrapolation bound. Interpretation must retain which physical or designed groups actually contribute dangerous screened candidates.

## 5.4 Use family viability and observations as diagnostic follow-up

The battery returns a viable family set after coordinate-specific testing. On the recorded synthetic family truths, containment is approximately .93-.95; on out-of-family alternatives, Table 10 identifies structures that escalate and structures that retain a wrong family. This supplies a filtering stage for further model comparison, with overlap explicitly recorded.

The World Bank and engineering suites extend the software and diagnostic record to observed data. Their agreement and divergence patterns can guide follow-up measurements or alternative models. Without known form labels or controlled interventions, those patterns supply compatibility evidence. The strongest empirical performance conclusions remain those supported by the ground-truth suites.

# 6 Scope and validation priorities

## 6.1 Scope of the completed evidence

The main experiments assess a fixed power-law candidate family on positive data. They do not evaluate the expression distribution of a search algorithm. S4 extends the coordinate construction to four linearizable families; the new policies add a quadratic rival and one frozen RBF rival. Strongly dependent errors, additive noise, other rivals, and application-specific tolerances need their own evidence. Residual permutation remains an empirical approximation under the evaluated conditions.

The historical S2 detection and gate-only pools contain 18 eligible H1 equations, and its endpoint-release pool contains 17. That outcome uses a sample-dependent first-coordinate probe and remains labeled as historical. P0-1 v2 retains known equation identities while changing sampling and calibration. P0-2 fixes domains and keeps structural groups together, but its primary confirmation pool deliberately contains both source-derived physical tasks and designed mechanisms. The number of formulas or simulation replicates is not the number of independent physical mechanisms. The additional amplitude input in designed controls also changes the population that clears $R^2$; their screened-dangerous frequency is not a physical prevalence estimate.

Source sampling metadata also require a physical interpretation audit. A positive finite mathematical region does not by itself establish all physical admissibility conditions or unit/type semantics. The source description of I.39.22 treats n as an integer, whereas its benchmark sampler is continuous. This is a historical development task, and its frozen P0-2 records are retained under the registered sampling interpretation. It is not one of the six physical-source confirmation tasks. The future source census flags such contract conflicts before any new physical sampling; the current study does not claim that every legacy domain has completed that broader audit.

The physical-source pool has no screened-dangerous primary H1 record in any of the twelve registered strata. Its corresponding conditional release rate is undefined throughout. The complete registered workflow nevertheless releases 0 of 3,600 observed dangerous physical-source fits across the grid: all are stopped by the $R^2$ screen. This overall secondary result describes successful filtering of these particular cases, with no gate interception required. The missing evidence concerns high-fitting physical candidates that are dangerous over the external region; it does not make every physical workflow outcome unmeasurable or establish physical-population safety. The p95 secondary endpoint contributes primarily one source group, with two additional records from a second group in one stratum. It remains secondary and cannot establish a broad physical policy ranking. The stronger multi-group primary comparison concerns the designed mechanisms.

All H0 tasks share the monomial family; their held-out grouping tests sampling-design transfer. Test H0 withholding is measured rather than guaranteed by the reference target. Intervals condition on the calibration-design identities, empirical calibration-round bank, and realized test data; they omit inner test-record resampling and uncertainty from drawing a different physical library. Ten rounds give an empirical sensitivity description, not a precise estimate of the full distribution of possible recalibrations. Finite-grid mean or maximum error is not a continuum guarantee, and training samples at different n are independent rather than nested.

## 6.2 Outcome-independent plan to expand physical coverage

A separate physical-coverage design, p03-physical-coverage-source-census-v1, was developed after P0-2 stage-1 results were available, while the registered full grid was being completed. It is not a retrospective preregistration of the current results. Its admission procedure reads the complete pinned 120-equation source catalog, source sampling and constraints, structural exposure identities, and validity certificates. It forbids use of P0-1/ P0-2 $R^2$ values, gate statistics, error or danger labels, screen-passing rates, or method rankings to admit a formula or choose its domain. The source count is a census, not a selected set of successful tasks. The scientific dimension limit is eight inputs; eight is also the maximum in this catalog, so that limit excludes no current source.

One common domain rule conditions the original sampling support on positive inputs and takes its central 5th-95th percentile box in the original sampling coordinate. Original integer support and sign branches are preserved before conditioning. The complete outer box must satisfy source constraints and strictly positive finite signed truth. Constants, coordinates, supplied unit metadata, and the source expression remain intact; absolute values, sign flips, output shifts, hand-picked positive lobes, and equation-specific domain rescue are forbidden. Missing unit or source-contract evidence is recorded explicitly, not inferred from variable names. Failure to establish a whole-box certificate is an unresolved obligation, not proof that the equation has no valid physical region anywhere.

The census records 86 mathematically admissible equations under the registered box rule (35 H0, 51 H1), 31 with unresolved validity or source obligations, two with nonpositive target throughout the box, and one without original positive-input support. Mathematical admission is separate from executable evaluation: eight eligible equations have more than four inputs and require adapter validation; two Gaussian laws need stable log-truth treatment rather than exclusion for floating-point underflow. These counts do not imply that physical/unit provenance and numerical checks are complete. Physical source-contract review remains independently pending for all 120 rows. A uniform textual-type scan finds seven documented integer-to-real relaxations and flags I.39.22 as source_type_review_pending because no such relaxation agreement resolves its integer description and continuous sampler.

The primary grouping conservatively merges variable renamings, global and separable monomial prefactors, fixed output powers or reciprocals, source variants, and registered mechanism families. Exposure propagates through connected components. It yields 13 components across the catalog; the eligible H1 equations occupy only four already exposed groups under this conservative rule. The grouping is coarser than P0-2's structural grouping. Counts from the two protocols describe their registered resolutions, not an objective number of physical mechanisms, and are not directly comparable. There are therefore zero undisputed prospective eligible H1 groups in this catalog under the primary grouping. A finer 14-component description leaves bonus.16 as one disputed combination candidate, retained in conservative development. Neither counting more source equations nor treating that disputed case as many mechanisms supplies independent physical confirmation.

The proposed future factorial crosses centered training widths 25%, 50%, and 75% of each fixed source-coordinate outer span with all four log-noise SDs $\{0.005, 0.01, 0.03, 0.05\}$ and $n \in \{200, 1000\}$, using 100 test replicates per complete cell. Every admitted group and variant uses the same design; difficult screens and numerical declines remain in the operational denominators. The sample-size choice is a structural feasibility rule: at the maximum eight inputs, a full quadratic has 45 columns and five-fold training at n=200 has 160 rows. It is not a result-based power selection. $R^2 \geq 0.99$ is only a downstream

release screen, never a source-admission or stopping rule.

This is a metadata-level planning registration with status pre_execution_registration_incomplete, not an executable experimental freeze. Exact task roles, source/physical certificates, stable log-truth and higher-dimensional adapters, width-specific fixed grids/weights, dimension-matched monomial calibration controls, source snapshots, and new seed namespaces must be completed before sampling. A new external physical-source catalog has not yet been chosen; its complete census and group-exposure review are prerequisites for multi-mechanism prospective confirmation. The present registration has generated zero new physical study samples, candidate fits, or diagnostic outcomes. It contributes a reviewable inclusion/exclusion procedure, not a completed extension of the physical empirical evidence.

### 6.3 Extension to discovered candidates and engineering decisions

The next algorithmic test must evaluate expressions actually selected by a discovery workflow. Candidate generation and selection must be fixed before independent diagnostic and external prediction evaluation. The null attached to the diagnostic must correspond to the selected candidate; a power-family diagnostic applied after search is not a general candidate-specific validity certificate. The exploratory PySR counterpart in Appendix H does not replace that experiment.

The observational suites show software applicability and diagnostic compatibility patterns, while their true equations and external loss are unknown. Engineering recommendations require reference outcomes over an intended operating domain, relevant error dependence, and an application loss tolerance. Those requirements go beyond detecting one form of statistical misspecification.

## 7 Reproducibility statement

The local source tree D:/form_check contains generators, recorded configurations, results, failure labels, and earlier analyses. The R2 manuscript, PDF, figures, and audit artifacts are preserved. The R3 manuscript integrates new evidence without replacing the historical experimental files. Appendix A reports original S1-S6 fitted-run totals; S7 reuses existing records. New P0-1/P0-2 dataset totals are additional and are not mixed into those historical counts.

Historical aggregation evidence is retained in the R2 bundle, including evidence/size.json, results.json, s7.json, task_intervals.json, and the record-level provenance audits. Figures D.1-D.5 describe those historical runs; Figure D.5's timing cannot be used as a timing comparison for the new RBF policy or updated inference paths.

P0-1 v1 and v2 reside in runs/p01_pilot_v1 and runs/p01_v2. The v2 frozen protocol, manifest, source snapshot, raw_policy.parquet, reference scales, policy thresholds, per-task counts, round summaries, and audit.json make fresh-sample calibration and uncertainty reconstructible. Source code is under experiments/release_validation; validation/audit_release_policy_v2.py independently recomputes policy metrics and uncertainty without importing their experimental implementation.

P0-2 resides in runs/p02_fixed_domain. protocol_frozen.json, task_registry.json, group_isolation.json, grid_manifest.json, and grids/ record the pre-sample registry and fixed truth endpoints; manifest and source snapshots bind the run to its implementation. Stage 1 and stage 2 retain their separate raw records,

checkpoints, decisions, thresholds, and summaries. The complete-grid aggregation combines the two nonoverlapping stage grids and does not replace either stage. The independent fixed-domain auditor reconstructs grid error from saved coefficients/truth and policy/count/uncertainty outputs from recorded raw diagnostic statistics; it does not independently recompute every A-G statistic from original training arrays. Registered secondary summaries retain their own endpoints and are not substituted for the primary outcome.

The proposed physical expansion is documented in docs/PHYSICAL_COVERAGE_PROTOCOL.md and experiments/physical_coverage_v1/protocol.json. The complete 120-row ledger, structural components, rules snapshot, and review hashes are retained under runs/p03_physical_protocol. Its manifest states incomplete pre-execution registration and zero generated study outcomes. An independent anti-selection/source-fidelity review passes for this snapshot; it checks the full census, uniform domain rule, and exposure metadata, without certifying all interval bounds or physical/unit/type contracts. Hashing this planning snapshot does not certify that the future study is runnable or that its pending physical coverage has been achieved.

Preservation audits compare full file-tree hashes for existing runs. Seed and recorded dataset-hash checks separate the new studies from the preserved P0-1 runs and from one another; source-file integrity is distinct from reconstructing every historical dataset lacking a saved data hash. Reproduction of fitted experiments is also distinct from reaggregation of saved raw statistics.

The original R2 artifact identifies repository https://github.com/leokynlang/form_check and historical data tag data-2026-10-18. This revision does not claim that its new local runs have been published in that historical release. The local manifests are the provenance record for the new evidence. Third-party redistribution decisions remain governed by data_release/ THIRD_PARTY_DATA.md and dataset_sources.json; copyrighted NIST WebBook SRD 69 source data remain source-download-only. No new external references or external empirical claims are added here.

The R3 bundle distributed with this manuscript contains the summary evidence for the new studies: revision_facts.json, per-task count and policy-threshold tables, round-level calibration summaries, paired and interval tables, and the independent audit outputs. The complete P0-1 v2 and P0-2 runs are larger and remain in runs/p01_v2 and runs/p02_fixed_domain. In P0-1 v2, decisions.parquet and raw_policy.parquet alone total about 41 MB, and the run also stores protocol_frozen.json, manifest.json, source_snapshot.json, the per-cell files, and audit.json. P0-2 keeps two nonoverlapping stage trees, a 1.8 MB grid_manifest.json, and freeze_independent_audit.json. A gitignore rule excludes record-level parquet data from the repository, so these files, including the 58,112 P0-1 v2 datasets and the 246,240 P0-2 datasets, ship as the release assets p01-v2-records.zip and p02-fixed-domain-records.zip on tag r3-2026-10-18. Each archive records a SHA-256 and byte count per member, and data_release/manifest_r3.json lists every member path and hash. Restoring them with tools/download_data.py --manifest data_release/manifest_r3.json makes the fitted experiments re-runnable from their original training observations. Without them, the summaries, protocols, source snapshots and audit scripts can be re-aggregated and checked, but the A-G statistics cannot be recomputed from those observations.

# 8 Conclusion

Specification policies can remove dangerous candidates while leaving an interpretable residual release channel. The historical benchmark maps directional blind spots, records false alarms, and shows why stable numerical inference matters. Its native-panel single-probe result remains informative but concentrated in two equations.

New validation separates H0 reference fitting, independent threshold calibration, and testing. The v2 affine score removes artificial CDF tail saturation, and repeated recalibration reveals why a stable decision rate can coexist with wide task-composition uncertainty. The fixed-domain study then evaluates the same fitted models over pre-sample external grids and holds related formulas inside structural groups.

The complete registered n/noise grid shows substantial residual release on the controlled mechanisms even at n=1000 and larger noise. F has the lowest primary release point in eleven strata, with an all-policy zero tie in the remaining stratum, while actual independent-test H0 costs differ. A larger panel therefore does not guarantee lower dangerous release at a common calibration budget in this library.

The physical-source result has a different boundary. Its observed dangerous fits are all stopped by $R^2$ screening, so overall dangerous release is zero in those cases; no screened-dangerous case remains to identify the gates' conditional physical performance. Correct-model losses are substantial at the larger noise levels, largely through the fit screen. The new source-census protocol records this coverage problem without selecting formulas or narrowing domains from observed performance, but its mathematical admission ledger and pending source contracts are not completed multi-mechanism physical confirmation.

The practical contribution is a reproducible way to compare complete decision policies using actual H0 costs, correct-model retention, dangerous-release denominators, and group/calibration uncertainty together. Broader physical coverage and independent validation after expression selection remain necessary before translating these controlled comparisons into general scientific discovery or engineering recommendations.

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# Appendix A - Suite composition

| Suite | Cells | Fits per cell | Total fits | Extra evaluations |
|---|---|---|---|---|
| S1 synthetic | 285 | 500 | 142,500 | - |
| S2 SRSD-Feynman | 480 | 500 | 240,000 | 240,000 test sets; 232,500 extrapolation probes |
| S3 family-stratified | 392 | 30 | 11,760 | - |
| S4 battery | 188 | 500 | 94,000 | - |
| S5 gate comparison (v3) | 102 | 200 | 20,400 | - |
| S6 real data + probe (v3) | 33 | 5 | 165 | - |
| **Total** | **1,480** | - | **508,825** | **472,500** |

Outside the seeded totals: the Appendix E engineering bench contributes 150 single-realization smoke cells (147 run), one fit per gate per cell.

# Appendix B - Failure registration taxonomy

| Reason | Meaning | Exercised by |
|---|---|---|
| `degenerate_zero_residual` | Zero-noise cell on exact ground truth; test undefined | S1/S2 $\sigma = 0$ cells |
| `rank_deficient` | Design matrix numerically rank-deficient | collinear designs |
| `insufficient_df` | $n - k < 1$ after augmentation | $n = 50$ multi-feature cells |
| `hc3_leverage_degenerate` / `hc3_singular_covariance` | HC3 undefined | degenerate built columns |
| `statistic_mismatch` | Wrapper statistic ≠ native statistic (drift sentinel) | continuous audit |
| `checked_false_*` | Native test declines to run (non-positive data, $n < 12$, non-finite) | not-applicable tasks |
| `runner_error` / generation failure | S5 SRSD-side generation exceptions | S5 |
| indicator archived / no data | S6 indicator not published for a year | S6 (`EN.ATM.CO2E.KT` → AR5 replacement, logged) |

All registered failures remain in the audit trail; denominators are documented per table. The battery's 94,000 and S5's 20,400 datasets contain zero data-layer failures.

# Appendix C - Statistical uncertainty and eligible populations

Tables 1 and 2 contain the fit-level Wilson 95% intervals recomputed by audit_size.py, with 6,000 and 26,000 valid positive-noise H0 evaluations per n, respectively. The S1 zero-noise declines are excluded from size estimation. These intervals describe simulation precision on the fixed scenario/equation pools, not cross-equation generalization.

At S2 n=200, the equation-level analysis in audit_results.py resamples the 13 H0 equations 10,000 times, with 2,000 records per equation. For C, the mean is .053115 and the 95% task-bootstrap interval is [.050308, .056385]. For D, the mean is .056385 and the interval is [.054615, .058385]. Leave-one-equation-out ranges are [.052000, .053542] for C and [.055625, .056667] for D. Section 4.1 uses the same results with rounded presentation. A lower bound displayed as .050 is not proof of exact coverage of .05.

The paired D-A intervals in Section 4.2 instead resample 18 H1 equations, preserving each equation's paired difference. They must not be substituted for fit-level binomial intervals, or for results on a 19-task pool. Pointwise Figure D.3 intervals and paired differences address different quantities.

The joint-release ratios have a separate task-level analysis. audit_task_intervals.py resamples the 17 eligible H1 equations (B=10,000) and recomputes pooled Ld/S, Ld/Sd, and Ld/Da from each task's counts. With releases confined to two equations, the pooled Ld/Sd of 7.5% has a 95% interval of [0%, 49.9%], and 12% of resamples contain no release. This interval describes between- equation heterogeneity; it is not a binomial interval on 34,000 records and does not include threshold-estimation uncertainty.

Seeded computational repeats within one equation do not create additional independent laws. Macro and record-weighted averages coincide only where the task design is balanced. The old pinv-path uncertainty results are historical numerical-path diagnostics, not current method-performance intervals.

# Appendix D - Result figures

Figures D.1-D.4 use the eligible record pools documented in audit_size.py and audit_results.py. Figure D.5 reports the historical timing measurements; its inference-path scope is specified in the caption. Figures D.6-D.7 report the separately frozen, independently calibrated fixed-region P0-2 evaluation; their task allocation and endpoints differ from historical S2.

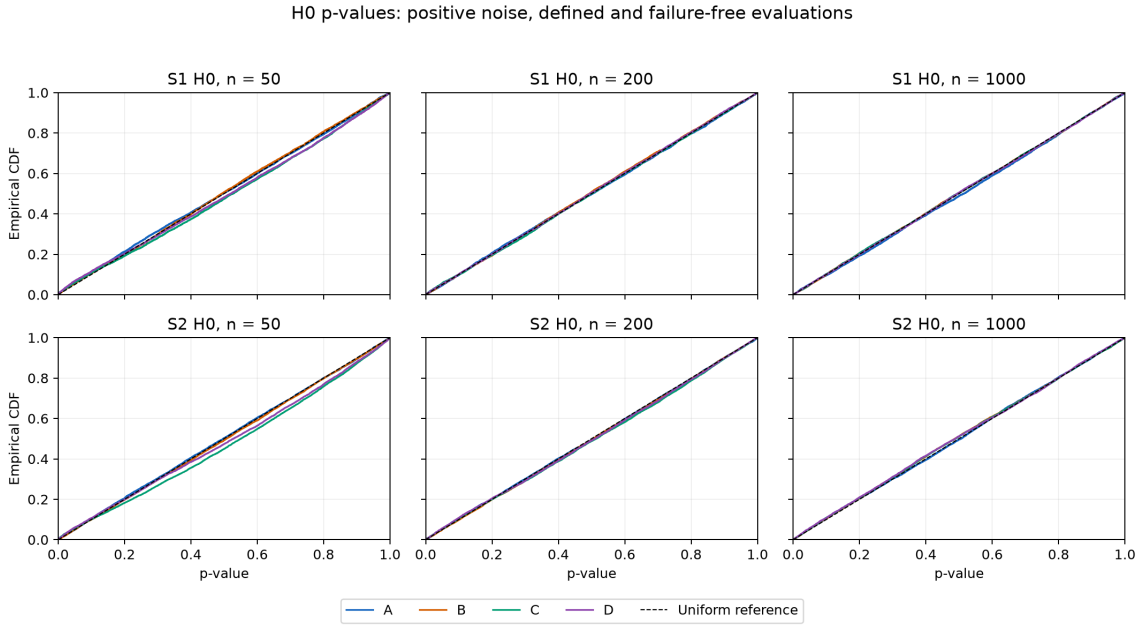



Figure D.1 - Empirical CDF of p-values under strict monomial H0 (uniform = calibrated): S1 synthetic H0 (left) and S2 SRSD-Feynman H0 (right). Approximate diagonal agreement does not establish exact 5% calibration; Table 2 reports the measured excess for C/D.

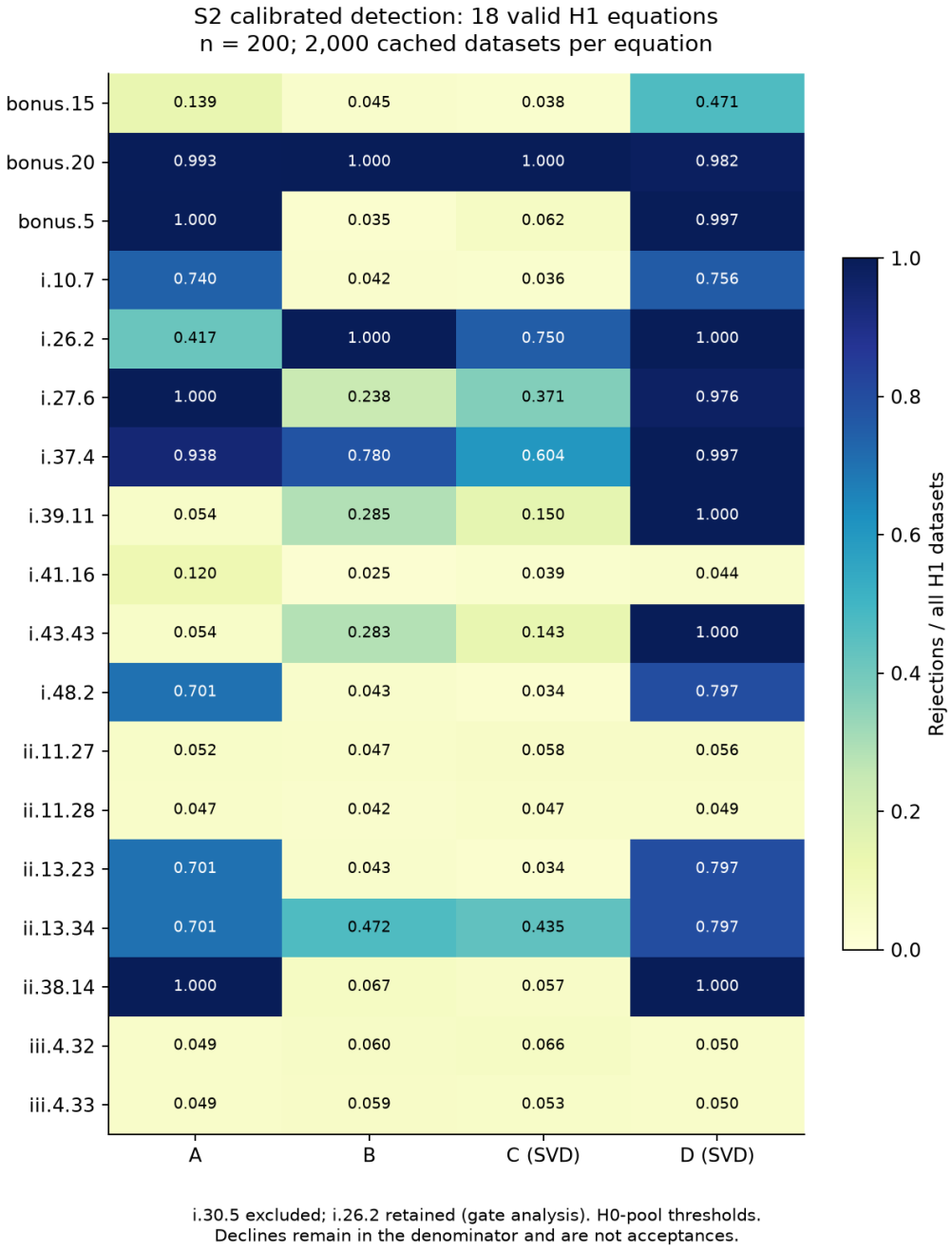



Figure D.2 - S2 H1 calibrated power per task (H0-pool 5% threshold, n = 200, rel > 0): one row per task, one column per gate; 18 sentinel-valid H1 tasks, excluding i.30.5. The two extrapolation-blind equations of §4.5 (i.39.11, i.43.43) are dark under A and B but reach 1.00 under D; iii.4.32 / iii.4.33 stay dark for every gate.

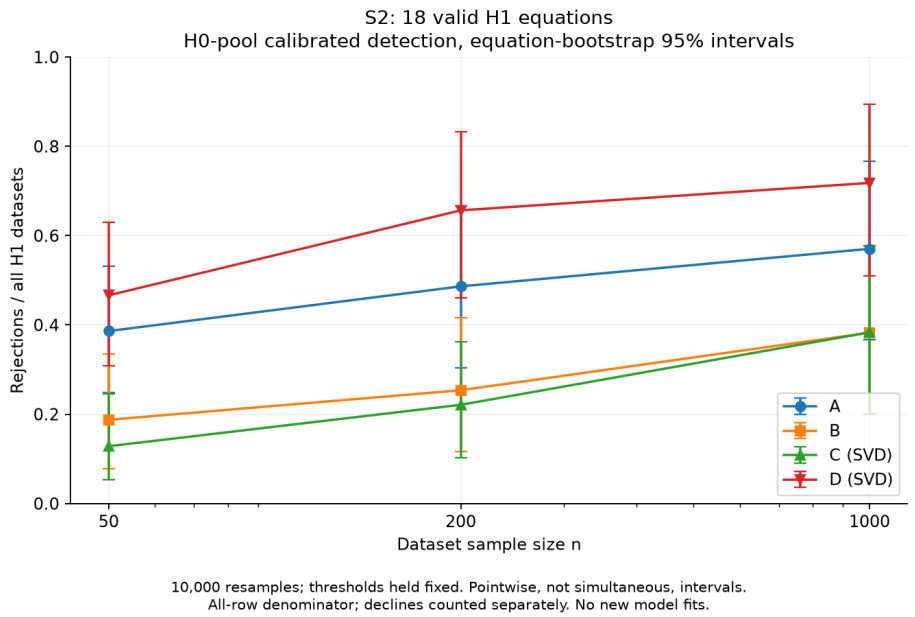



Figure D.3 - S2 H1 calibrated power versus n, with task-level bootstrap 95% intervals (10,000 replicates over the 18-task pool). Declines stay in the denominator but are not rejections. D has the largest mean in this pool; paired D-A intervals are in Section 4.2.

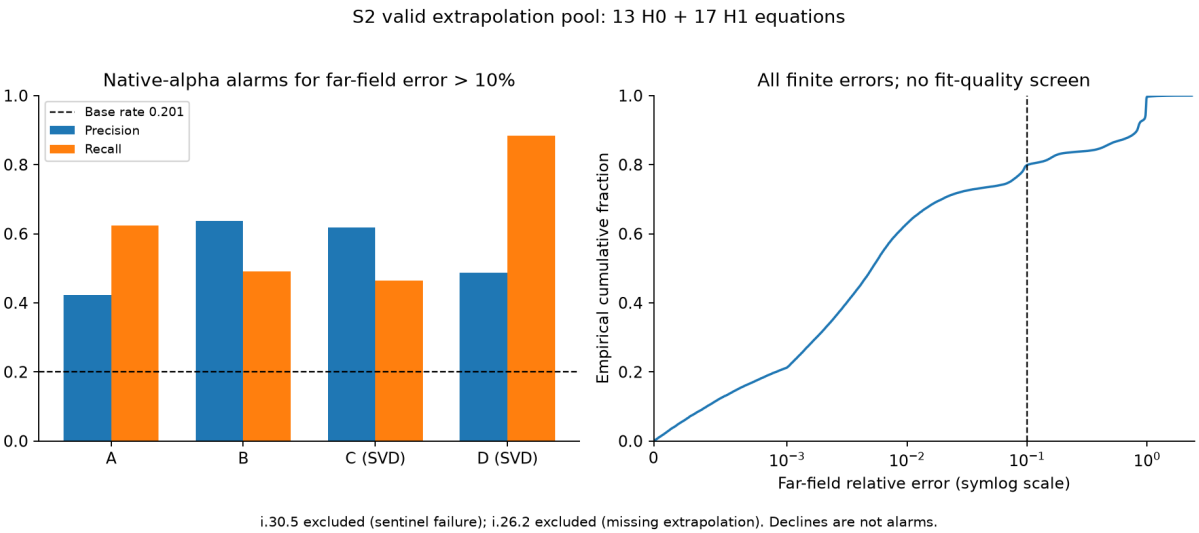



Figure D.4 - Left: alarm precision and recall for far-field error >10% on the revised S2 evaluable pool (13 H0 and 17 H1 tasks, positive noise, all n). Right: the corresponding error distribution. The task and missing-output filters match Section 4.10.

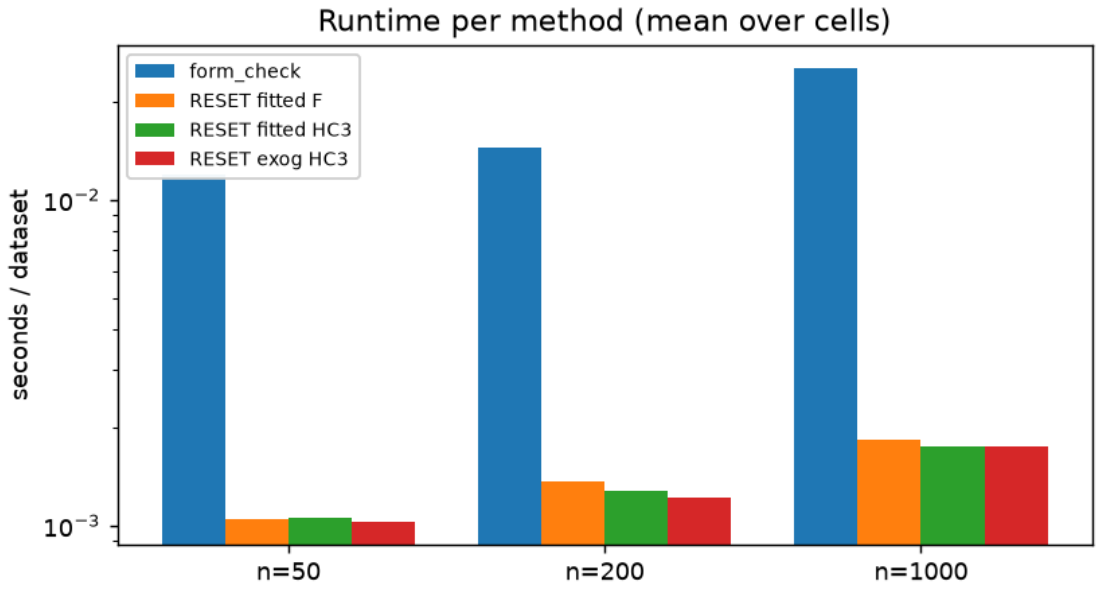



Figure D.5 - Per-dataset wall time; all four tests stay under 30 ms across n in the recorded historical S1 run. These timings do not establish equal runtime for the revised SVD inference path.

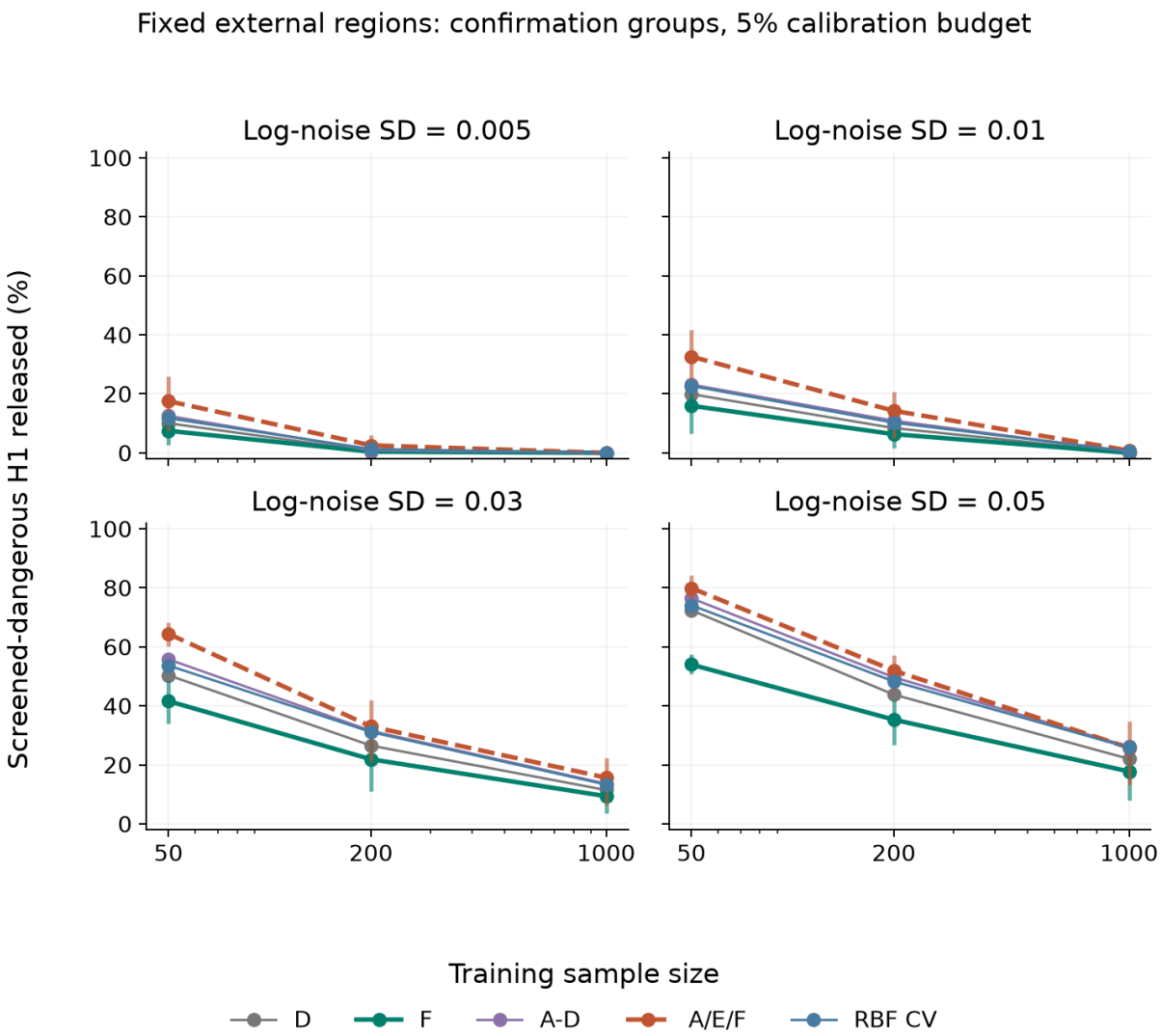



Figure D.6 - Dangerous H1 release after the fit screen on all 12 registered P0-2 sample-size/noise strata, at a common 5% calibration withholding budget. Vertical intervals for F and A/E/F are paired structure/design-group by calibration-round percentile intervals; all policy intervals and undefined denominators remain in the released summaries. Intervals condition on the fixed calibration design pool and realized test bank. Equal calibration budgets do not imply equal actual test H0 withholding.

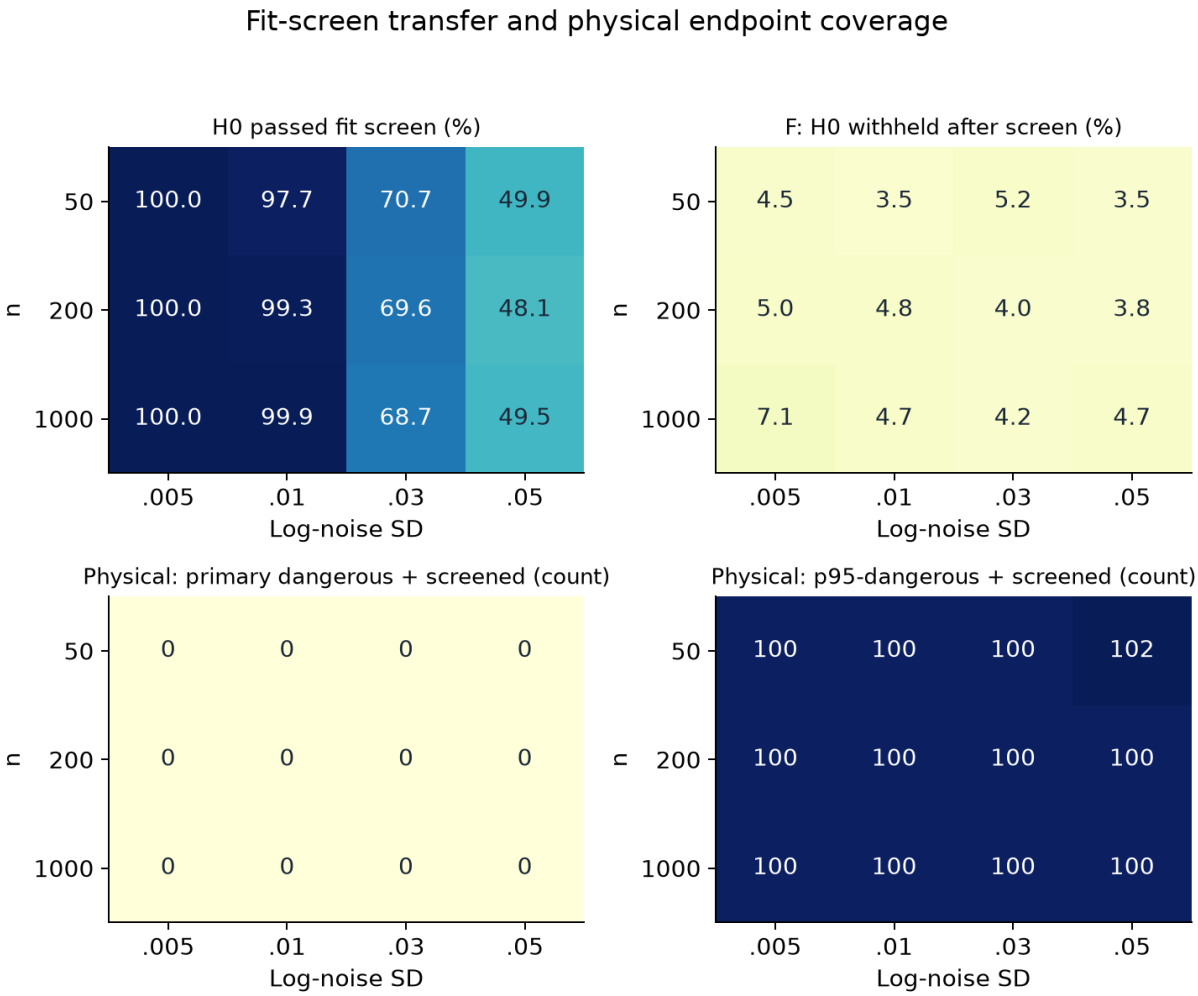



Figure D.7 - P0-2 fit-screen transfer and physical endpoint coverage. Top: confirmation H0 screen retention and F's test withholding conditional on screening. Bottom: physical-source records that are both screened and dangerous under the primary mean-error and registered secondary p95-error outcomes. A zero danger denominator makes the corresponding dangerous-release rate undefined, not zero. Counts do not establish independent physical-group coverage.

# Appendix E - The real-engineering data bench (companion to S6)

S6 probes mechanism transfer on macroeconomic aggregates. The deployment domain these gates are actually built for is physical-engineering measurement - fluid, civil, hydraulic, energy, meteorological, seismic, chemical, and aerospace data, where power laws are the daily candidate models. This appendix documents a companion bench of that kind, run under the same contract as S6: real data has no ground truth, so the bench registers compatibility and divergence, not error rates; its cells are single realizations and stand outside Appendix A's seeded totals.

**Composition and discipline.** Nine open sources with source-specific licenses; 91 datasets; 150 candidate log-log relations registered in the local manifest with per-relation physical notes; 761,515 raw rows (Table E1). Raw bytes are cached (~115 MB) with SHA-256 hashes; cleaning is replayable and imputation-free - rows with missing or non-positive values are dropped and logged, and the few unit conversions are explicit in the manifest (e.g., Whiteside °C → K). Simulator datasets that violate the real-data discipline (a

marine-engine and two wave-energy numerical models) are excluded and listed, as are WebBook fluid pages below the 10-row rejection-detection floor; one gauge whose instantaneous series was never published is logged as an empty skip. ERA5 cells are marked as reanalysis, not direct measurement.

**Table E1 - Composition by source.**

| Source | Datasets | Relations | Rows | Examples |
|---|---|---|---|---|
| USGS NWIS rating curves | 12 | 12 | 442,269 | discharge ~ gage height, 12 gauges |
| UCI | 11 | 38 | 145,920 | CCPP, Tetouan power, steel, yacht, concrete |
| EPA fueleconomy | 1 | 4 | 48,624 | $CO_2$ ~ MPG; combined MPG ~ displacement |
| Open-Meteo / ERA5 | 10 | 20 | 87,600 | 100 m ~ 10 m wind speed (shear); gust factor |
| OpenML | 3 | 8 | 21,997 | aileron position; CPU utilization |
| USGS FDSN | 1 | 3 | 12,334 | felt reports, stations, intensity ~ magnitude |
| RDatasets | 14 | 26 | 1,349 | cars, trees, puromycin, stormer, rock |
| NIST StRD | 13 | 13 | 968 | certified nonlinear calibration curves |
| NIST WebBook | 26 | 26 | 454 | phase-transition enthalpies ~ temperature |
| **Total** | **91** | **150** | **761,515** | - |

**Protocol.** Each cell applies the §3.2 gates unchanged at $\alpha = 0.05$ - A with its $B = 999$ permutation p, B/C/D as registered - on the positivity-filtered relation, subsampled to $n \leq 800$ (seed 42) to bound permutation cost. The run is a smoke pass: one realization per cell, compatibility plus first verdicts, not a rate estimate. Of 150 cells, 147 run; three are registered `insufficient_n` by design (Bennett5 and Roszman1 have no positive rows after the log-space positivity filter; Danwood has $n = 6$) - the bench's honest-failure probes working as intended.

**Table E2 - Smoke verdicts on the 147 run cells.**

| Gate | Rejections | Registered failures |
|---|---|---|
| A - permutation curvature | 97/147 (.660) | 10 declines, all `checked_false_small_n` ($n \leq 11$) |
| B - RESET fitted F | 111/147 (.755) | 0 |
| C - RESET fitted HC3 | 108/147 (.735) | 0 |
| D - RESET exog HC3 | 94/147 (.639) | 3: `insufficient_df` ($n = 13$, four features), `hc3_leverage_degenerate` ($n = 546$), `rank_deficient` |

A and B agree on 123/147 cells (92 both reject, 31 both quiet; 5 A-only, 19 B-only). The failure columns are themselves informative: D's three declines are exactly the three HC3 failure modes of Table 2 and Appendix B, now on observational data; A's ten are uniform declines-to-run on $n \leq 11$ cells; B and C never fail. Nothing fails silently - every declined evaluation is registered with a typed reason.

**Designed regimes.** The bench's design document assigns expected regimes to 66 of the 150 cells before the smoke pass (Table E3). The assignment is a reading aid, not a scored test: the H0-side laws are approximate by design, and at n up to 800 an approximate law is a detectable null. The 12 rating-curve

cells - whose registered form Q ~ h omits the documented stage offset $h_0$ of the true rating law $Q = C(h - h_0)^n$, a mismatch noted in the manifest - are rejected unanimously by A and B alike (12/12): detection of a documented misspecification, not a false alarm. The five H0-side cells A retains are those with no designed departure mechanism - the braking-distance $v^2$ law, the tree allometry, the road-load relation - plus the two shear sites with the smallest fitted curvature.

**Table E3 - Designed-regime cells, rejections at α = 0.05.**

| Designed regime | Cells | A | B | C | D |
|---|---|---|---|---|---|
| Approximate power laws (H0-side) | 30 | 25 | 27 | 23 | 23 |
| Known non-power-law physics (H1-side) | 36 | 20 | 24 | 24 | 14 |

(A's H1-side count includes its nine WebBook declines at n ≤ 11 as non-rejections.) On the H1 side no gate sweeps the board: the misses concentrate in WebBook enthalpy cells whose measured temperature window keeps the local power-law approximation good - e.g., toluene Δ_vapH ~ T: all four gates quiet at n = 26 - a mutual-miss cell recorded as measured.

**Table E4 - Named cells: mechanism echoes and divergences; p-values at α = 0.05 in A / B / C / D order, - = declined with registered reason.**

| Cell | A / B / C / D | Reading |
|---|---|---|
| Wind shear $v_{100} \sim v_{10}$, 10 ERA5 sites (n = 800 ea.) | 8/10 / 10/10 / 7/10 / 7/10 | B flags every site ($p \leq 2\times10^{-4}$); A retains exactly the two smallest-curvature sites (.23, .41) - a graded response, not a threshold artifact |
| Gust factor ~ $v_{10}$, same sites (n = 800 ea.) | 10/10 ×4 | unanimous rejection at every site; no truth to adjudicate |
| puromycin rate ~ conc (Michaelis-Menten), n = 23 | **.011** / .069 / .169 / .169 | A detects the saturation curvature, B misses at α = .05 - Table 3's small-sample power gap, live |
| stormer Time ~ Viscosity + Wt (calibration law), n = 23 | .001 / $7\times10^{-4}$ / $9\times10^{-3}$ / - | single-feature cells quiet (A .94, B .99); the second feature carries the interaction the law contains |
| EPA $CO_2$ ~ MPG (stoichiometric inverse, exponent −1), n = 800 | .030 / $1.5\times10^{-4}$ / .727 / .727 | A and B reject on a near-exact physical law bent by fleet drive-mix curvature |
| EPA combined MPG ~ displacement (road load), n = 800 | .386 / .596 / .599 / .599 | designed power law passes |
| CCPP PE ~ V, n = 800 | .642 / $8\times10^{-4}$ / ≈0 / .023 | proxy curvature: V correlates .81 with the strongly curved AT, so B/C's fitted-power columns inherit AT's curvature while A's V-only direction set cannot - the mirror of dilution |
| automobile horsepower ~ engine size, n = 203 | .878 / $1.2\times10^{-4}$ / .094 / .094 | divergence under heteroskedastic body-style scatter |
| rock perm ~ area + peri, n = 48 | .835 / $1\times10^{-5}$ / $5.6\times10^{-6}$ / .596 | A/D quiet, B/C reject; no truth to adjudicate |
| housing price ~ 4 features, n = 546 | .723 / .683 / ≈0 / - | D's leverage-degeneracy failure mode, live at n = 546, while A/B/C return verdicts |

**Scope.** What the bench supports is what S6 supports: compatibility at scale (147/150 cells run across nine sources, zero silent failures), failure registration on observational data, and mechanism-echo cells whose directions match the synthetic map. Its per-gate rejection totals (97/111/108/94 of 147) are compatibility statistics on single realizations without ground truth - they rank nothing, and reading them as performance would repeat exactly the leaderboard confusion the benchmark exists to correct - A's observed non-rejections (the shear sites, the road-load law, puromycin at n = 23), A's interaction and dilution behavior (stormer, CCPP), D's structural fragility (all three failure modes live). What it cannot support: rates, calibration, or rankings - single realizations, no ground truth, and time-ordered NWIS/Open-Meteo series whose permutation-independence assumption is an approximation marked in the manifest. All four p-values per cell are recorded locally (`real_data/smoke_results.csv`) with the manifest (`real_data/clean/manifest.json`, SHA-256 per source) and replayable fetch/prepare/smoke scripts.

# Appendix F - Core code listings

Abridged reference implementations (error handling and instrumentation omitted; full implementations are present in the local benchmark project). Listing 1 is the wrapper around the project's native `form_check`: it recomputes the curvature statistic in log-log space, verifies it against the native value (drift sentinel), and derives the empirical p-value from a vectorized residual-permutation null. Listing 2 is one battery member: the same statistic in a family's linearizing coordinates with a wild bootstrap null.

Listing 1 - Method A wrapper: permutation curvature test (common.py, abridged).

```
def form_check_with_p(data, target, features, seed, b_perm=999):   # (abridged)
    # Method A wrapper: statistic = coefficient of the mean-square column
    # in log-log space; null = residual permutation, vectorized via pinv row.
    fc = dimreflex_form_check(data, target=target, features=list(features),
                              n_boot=200, random_state=seed)
    if not fc.get("checked", False):
        return {..., "fail": classify_reason(fc["reason"])}

    lx = np.log(data[list(features)].values)
    ly = np.log(data[target].values)
    n = len(ly)
    quad = (lx[:, 0] - lx[:, 0].mean())**2 if len(features) == 1 \
           else ((lx - lx.mean(axis=0))**2).mean(axis=1)
    base = np.column_stack([np.ones(n), lx])
    design = np.column_stack([base, quad])

    coef, *_ = _ols(design, ly)
    c2 = float(coef[-1])                    # statistic: curvature c2
    if not np.isclose(c2, fc["curvature"], rtol=1e-6, atol=1e-10):
        return {..., "fail": "statistic_mismatch"}   # native-drift sentinel

    bcoef, *_ = _ols(base, ly)              # restricted fit -> perm null
    pred, resid = base @ bcoef, ly - base @ bcoef
    if float(resid @ resid) <= 1e-12 * float(ly @ ly):
        return {..., "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(int(seed) * 1_000_003 + 17)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in y
    c2_base = float(pinv_row @ pred)
    idx = np.argsort(rng.random((b_perm, n)), axis=1)
    null_c2 = c2_base + resid[idx] @ pinv_row    # B permutations at once
    mu, sd = float(null_c2.mean()), float(null_c2.std())
    p_emp = (1.0 + float(np.sum(np.abs(null_c2 - mu)
                                >= abs(c2 - mu)))) / (b_perm + 1.0)
    return {"p": float(p_emp), "rej": bool(p_emp < ALPHA),
            "t": float((c2 - mu) / sd)}
```

Listing 2 - Battery member: family-coordinate curvature test with wild bootstrap null (battery.py, abridged).

```
FAMILIES = {                                    # (u(x), v(y)) per family
    "power":       (np.log,          np.log),
    "exponential": (lambda x: x,     np.log),
    "logarithmic": (np.log,          lambda y: y),
    "rational":    (lambda x: 1.0/x, lambda y: 1.0/y),
}

def spec_test_wild(X, y, uf, vf, seed, b_wild=999):   # one battery member
    # H0: conditional mean linear in family coordinates (u, v).
    # Statistic: coefficient of quad = mean_j(U_j^2); null: wild bootstrap.
    U = _standardize_cols(uf(X))         # per-column transform + standardize
    v = vf(y)
    if (X <= 0).any() or (y <= 0).any():
        return {"p": np.nan, "fail": "nonpositive"}
    base = np.column_stack([np.ones(len(v)), U])
    quad = (U**2).mean(axis=1)
    design = np.column_stack([base, quad])
    coef, *_ = np.linalg.lstsq(design, v, rcond=None)
    t_obs = float(coef[-1])
    b, *_ = np.linalg.lstsq(base, v, rcond=None)
    resid = v - base @ b
    if float(resid @ resid) <= 1e-12 * float(v @ v):
        return {"p": np.nan, "fail": "degenerate_zero_residual"}

    rng = np.random.default_rng(seed)
    pinv_row = np.linalg.pinv(design)[-1, :]     # c2 is linear in v
    c2_base = float(pinv_row @ (base @ b))
    signs = rng.choice((-1.0, 1.0), size=(b_wild, len(v)))
    null = c2_base + (resid[None, :] * signs) @ pinv_row   # Rademacher wild
    mu, sd = float(null.mean()), float(null.std())
    p = (1.0 + float(np.sum(np.abs(null - mu) >= abs(t_obs - mu)))
         ) / (b_wild + 1.0)
    return {"p": float(p), "t": float((t_obs - mu) / sd)}
```

# Appendix G - Audit and correction history

Sections G.1-G.2 retain the historical correction record through R2. Statements about work performed in those revisions do not describe the additional fitted experiments in R3; the R3 additions are listed in Appendix I and Sections 3.8-3.9 and 4.12-4.13.

## G.1 Statistical, provenance, and presentation corrections

This ledger supersedes the mixed v2-v5 change notes in the supplied working draft. The original PDF and Markdown are preserved unchanged. Historical scripts and reports remain evidence of how earlier summaries were produced, not automatic authorities for the current numbers.

1. **Task pools and failures.** Record-level joins identify 7,500 S2 statistic-sentinel failures for feynman-i.30.5 over the full grid. Revised power and joint silence use 18 H1 tasks; joint release additionally requires a finite far-field outcome, leaving 17 H1 tasks because i.26.2 lacks that outcome. Original generated-fit totals are not relabeled as eligible observations.

2. **Joint-release accounting.** Table 15 now defines all four counts and three conditional rates on the same eligible records. Its pooled counts are A=65,992, B=6,403, C=481, D=36,169 (shown as S, Sd, Ld, Da in R2). Gate-only silence is not substituted for complete-pipeline release. Comparisons between different conditional fractions no longer support a claim of danger-concentrated failure.

3. **S7 provenance and gate sets.** S7 is a reanalysis of existing S2/S5 records, not new sampling. The S5 arm uses A/E/F, not A-F. A paired A versus A/E/F comparison replaces the invalid cross-arm interpretation. Likelihood ratios and equal-prior posteriors have separate formulas. The S5 cache's weaker historical sentinel provenance is disclosed.

4. **SVD power and calibration.** Table 3 and the paired bootstrap use one 18-task pool and explicit decline handling. D-A effects are recomputed from the SVD data, not obtained by reversing old signs. The n=200 H0 task-level intervals are consistent between Section 4.1 and Appendix C. The manuscript no longer describes all C/D Wilson intervals as covering .05. Table 1 is also corrected: the 1,500 zero-noise declines per n are excluded, leaving 6,000 evaluated H0 records, rather than diluting the rejection rate with undefined tests.

5. **Claims, references, and artifacts.** The introduction and related work distinguish existing structural-recovery evaluation from this paper's diagnostic-gate experiment. References [6] and [11] have corrected authorship. Sample-size effects are not attributed solely to screening. Local script presence is not described as a public archive or a completed independent replication. The original scripts are not modified.

6. **What was checked in this revision.** The accompanying audit_size.py, audit_results.py, and audit_s7.py re-aggregate existing caches, record input hashes and task filters, and generate machine-readable evidence. build_pdf.py renders the corrected manuscript and figures. No new model fitting, public release, or engineering safety validation was performed. Remaining research limitations are in Section 6.

7. **Research presentation revision (R2).** R2 reorganizes the manuscript around detection, false-alarm cost, and extrapolation-related joint release. The abstract and introduction state the three experimental contributions; the discussion presents method-selection lessons; Section 6 specifies two priority validation experiments and the conditional extension to discovery-generated candidates. Historical accounting corrections are retained in this appendix. All numerical data rows in Tables 1-16 and the companion engineering tables are preserved from R1. This revision adds no model fits or completed validation experiments. Appendix H reports an exploratory PySR counterpart outside the main tables.

8. **Review-response revision (R2, 2026-10-18).** The artifact is public at https://github.com/leokynlang/form_check with data release tag data-2026-10-18. Section 7 now points to it; reference [17]'s licensing description is corrected; Section 4.10 adds a pooled task-level threshold sweep; the abstract, introduction, discussion, and conclusion state the task concentration of the 481 releases; and Appendix H adds the exploratory PySR counterpart. No aggregate (S, Sd, Ld, Da), table row, or figure in Tables 1-16 changed.

## G.2 Numerical implementation audit

The implementation audit compares the original pinv(X′X) calculation, an SVD calculation using pinv(X), and statsmodels HC3-Wald with the same F reference. The exog-HC3 check spans all 13 H0 equations, stratified by equation, sample size, and noise level (176 fits on identical input data). Maximum absolute p-value difference between the SVD and statsmodels paths is .0023; maximum relative statistic difference is .0038; covariance Frobenius relative difference is at most $3\times10^{-4}$. These measurements establish close agreement on the tested fits.

The original path rejects at rate 1.00 on seven tasks whose measured cond(X′X) is at least $4\times10^{17}$, with measured cond(X) from $6.5\times10^{8}$ to $3.2\times10^{10}$. On these small audit samples, both stable paths have task rejection rates from 0 to .10. The two better-conditioned tasks ii.27.16 and ii.27.18, with cond(X) near 10, show no corresponding failure. A fitted-HC3 check on two tasks separately gives 8/120 rejections (.067, Wilson [.034, .126]).

Forming X′X squares the spectral condition number before the pseudoinverse cutoff is applied. On the observed near-collinear designs, discarded small singular directions alter the hat and sandwich calculations. The audit attributes the large original-path distortion to this numerical realization; it does not establish a failure of the HC3 correction. The SVD path supplies the C/D method comparisons reported in the main tables.

The earlier S2 H0 summaries for the original path were C=.417-.422 and D=.563-.565 across sample sizes. These historical values are withdrawn as evidence about HC3's statistical performance. The implementation audit was a separate, one-off comparison rather than a rolling software check on every fit. Close implementation agreement and empirical size are different validation questions.

# Appendix H - Exploratory PySR counterpart

This appendix reports an exploratory check that applies the four gates to expressions emitted by a symbolic-regression search. It is separate from the main benchmark: candidate generation is not frozen, the sample is small, and the counts are compatibility evidence rather than rate estimates. The data are in evidence/deep_analysis.json and runs/pysr_gate/wide_svd/ in the data bundle.

**Protocol.** PySR searched the log-space training design with a restricted arithmetic operator set (+, -) matching the power-law hypothesis class. Each emitted expression was refitted as a prespecified candidate and passed through gates A-D under the same native thresholds. The run produced 360 candidates across the evaluated tasks.

**Results.** All 360 candidates: 87 jointly silent (24.2%). In the $R^2 \geq 0.99$ layer (183 candidates), 70 are jointly silent (38.3%), close in magnitude to the 38.6% S2 four-gate figure at n=50 although the candidate pools differ. In the far-field error >10% layer (142 candidates), only 3 are jointly silent (2.1%); the $R^2 \geq 0.99$ and far-field >10% intersection contains 2 candidates, both rejected by gate D ($p \approx 5\times10^{-29}$) while A, B, and C are silent. Per-gate rejection over all candidates: A 182 (50.6%), B 103 (28.6%), C 82 (22.8%), D 189 (52.5%); A is the sole rejecting gate on 45 candidates and D on 41.

**Scope.** These counts describe one search configuration and 360 candidates, of which only two meet the joint danger criterion. They do not establish a pass rate for discovery algorithms, do not rank the gates,

and are not merged into Tables 1-16. Their purpose is to show that the same diagnostic record can be attached to search-selected expressions and that the main benchmark's failure channel has a counterpart in this pilot.

# Appendix I - New validation and proposed-work boundary

P0-1 v2 is a known-equation fresh-sample experiment; P0-2 is a prospectively fixed-domain and grouped-task experiment. The latter's physical-source and designed-mechanism labels remain separate in the artifact and in interpretation. A newly executed source formula does not become an independent mechanism merely because its coefficient, variable names, or positive restriction differs.

Stage 1's two strata and stage 2's ten strata are nonoverlapping parts of the same frozen 12-cell grid. Their joint raw record count is 246,240, with 108,000 independent test datasets. Reusing each test bank across calibration rounds changes decisions, not the independent sample size. The recorded B/C rank declines remain in development accounting; historical R2 tables and all prior run artifacts are preserved. The complete-grid summaries are derived outputs and do not replace either stage's raw results.

The proposed expansion of physical coverage is a separate future study. Its written design supplies no empirical evidence that more physical structures pass a fit screen, become dangerous externally, or change the relative gate ranking. Any later software or eligibility change must preserve this record and state its timing relative to outcome access.